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Original Article

# Evaluating pre-processing and deep learning methods in medical imaging: Combined effectiveness across multiple modalities ☆

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ABSTRACT

Medical imaging is critical in modern healthcare for accurately detecting and diagnosing various medical conditions. Advanced computational techniques, particularly preprocessing methods and deep learning models, have demonstrated significant potential for improving medical image analysis. However, determining the optimal combination of these techniques across different types of medical images remains a challenge. Using empirical experiments, this evaluation research investigates the effectiveness of five popular pairs of preprocessing techniques combined with five widely used deep learning models. Preprocessing methods evaluated include CLAHE + Butterworth, DWT + Threshold, CLAHE + median filter, Median-Mean Hybrid Filter, and Unsharp Masking + Bilateral Filter, concatenated with deep learning models: EfficiencyNet-B4, ResNet-50, DenseNet-169, VGG16 and MobileNetV2. The performance of these combinations was evaluated through experiments carried out on eight diverse and commonly used datasets encompassing various medical imaging modalities. These datasets include two X-ray collections: the COVID-19 Pneumonia Normal Chest PA Dataset and the Osteoporosis Knee X-ray Dataset; two CT scan datasets: the Chest CT-Scan Images Dataset and the Brain Stroke CT Image Dataset; two MRI datasets: the Breast Cancer Patients MRI and the Brain Tumor MRI Dataset; and two ultrasound datasets: the Ultrasound Breast Images for Breast Cancer and the MT Small Dataset. Our findings show that the Median-Mean Hybrid Filter and Unsharp Masking + Bilateral Filter are the most effective preprocessing methods, achieving an efficiency rate of 87.5%. Among the deep learning models, EfficiencyNet-B4 and MobileNetV2 are the highest performing models with an efficiency ratio of 75%, with MobileNetV2 providing up to 34% shorter runtime compared to other models. This study provides a thorough evaluation of the performance of different preprocessing methods and deep learning algorithms across commonly used medical imaging modalities. Presenting empirical results from our experiments offers practical insights into choosing the most suitable preprocessing techniques and deep learning models for various types of medical images. These findings are intended to support improvements in diagnostic accuracy and efficiency in medical imaging, offering a valuable reference for enhancing image-based diagnostic processes.

## 1. Introduction

Various medical imaging techniques such as Ultrasound, CT, X-ray, and MRI for diagnostic purposes are commonly used in the medical field today. These medical image processing methods have transformed medicine by providing a non-invasive means to observe and assess the body's internal structures and functions. These imaging diagnostics are crucial as they provide vital information that can lead to early detection of diseases, accurate diagnosis, and effective treatment planning. Medical image diagnostics enable healthcare professionals to monitor disease progression, assess the efficacy of treatments, and guide surgical interventions with precision, enhancing treatment success and lowering the risk of fatality.

Recent progress in deep learning (DL), particularly in Convolutional Neural Networks (CNNs), has significantly automated this process [1], such as extracting complex features from medical images, automatically classifying diseases through images, image-based segmentation to assist diagnosis, monitoring treatment progress, and making informed decisions quickly [2,3]. However, these models heavily rely on input data [4]; high-quality and accurately labeled datasets are crucial for

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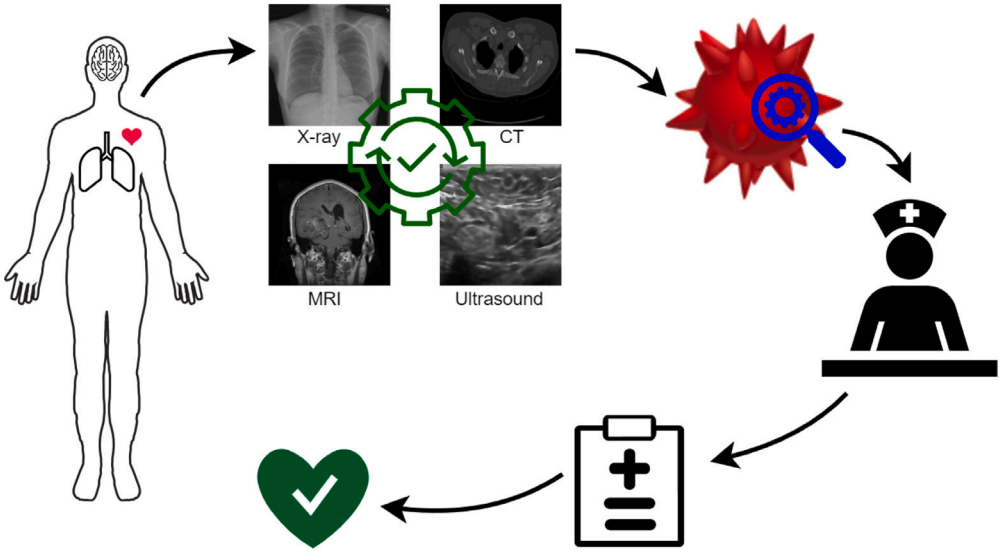



**Fig. 1.** The role of preprocessing and DL in medical imaging, from image enhancement to disease detection and treatment planning.

training effective CNNs. Biases within the data can lead to biased algorithms, potentially impacting diagnosis and treatment decisions. Input data are also affected by noise from creating these medical images, such as noise from devices, unrelated details in the background, uneven brightness distribution, inconsistent contrast, and artifacts introduced during the imaging process. These challenges are particularly common in medical imaging because of the intricate nature of the human body and the constraints of current imaging technologies. For example, X-ray images often suffer from noise caused by the scattering of X-rays, while MRI images can exhibit artifacts resulting from patient motion or magnetic field inhomogeneities. CT scans may be impacted by streak artifacts from dense materials and are also susceptible to additive white Gaussian noise (AWGN) [5]. Similarly, ultrasound images are also vulnerable to speckle noise caused by interference patterns, which makes the analysis and interpretation of these medical images more challenging [6,7].

Therefore, preprocessing images before applying them to model training is crucial to mitigate these challenges. Recently, there has been growing attention in the medical imaging field toward the use of preprocessing techniques. Texture analysis, which examines the structural patterns of image pixels, can provide insights into the internal composition of living tissues [8]. These methods aim to enhance image quality, reduce noise, and improve overall interpretability. Preprocessing techniques commonly used in medical image analysis include:

- Contrast Limited Adaptive Histogram Equalization (CLAHE) [9]: Enhances local contrast, particularly useful for improving the visibility of details in regions that are both bright and dark.
- Discrete Wavelet Transform (DWT) with Thresholding [10]: Breaks down the image into different frequency components, allowing noise removal by thresholding coefficients in the transformed domain.
- Butterworth Filter [11]: A type of frequency domain filter used for smoothing and noise reduction.
- Mean Filter [12]: A basic spatial domain filter that smooths noise by replacing each pixel's value with the average of its surrounding pixels.
- Median-Mean Hybrid Filter: Combines the advantages of mean and median filtering, balancing noise reduction and preservation of edges.
- Unsharp Masking [13]: It enhances contrast by sharpening edges, achieved through subtracting a blurred version of the image from the original.
- Bilateral Filter [14]: A non-linear filter that preserves edges while reducing noise, particularly effective for preserving details in medical images.

These preprocessing methods can be combined in various ways to match the unique characteristics of the medical images under analysis. These methods include CLAHE + Butterworth, DWT + Thresholding, CLAHE + Mean Filter, Median-Mean Hybrid Filter, and Unsharp Mask + Bilateral Filter. We systematically paired these methods with advanced CNN architectures such as EfficientNet-B4, ResNet-50, DenseNet-169, VGG16, and MobileNetV2 to determine the optimal model-preprocessing combinations for a variety of medical images. Fig. 1 illustrates the process of integrating preprocessing techniques with DL models in medical imaging. It begins with capturing images from modalities like X-ray, CT, MRI, and Ultrasound. Preprocessing techniques are subsequently used to boost image quality, minimize noise, and enhance clarity. These refined images are analyzed by DL models, which detect anomalies or diseases. The insights gained support medical professionals in diagnosing conditions and planning treatments, ultimately contributing to better patient outcomes.

***Research Gap:***

While there have been significant advancements in the use of deep learning (DL) for analyzing medical images, many studies overlook the crucial impact of preprocessing methods, which can significantly influence model performance. Much of the current research focuses predominantly on developing DL architectures, often without systematically evaluating the preprocessing methods applied prior to training. Similarly, some efforts emphasize the development of preprocessing techniques but fail to assess their impact across different models and imaging modalities. The absence of thorough studies examining the interplay between different preprocessing techniques and deep learning models across various medical imaging modalities hinders a deeper understanding of how the best preprocessing choices can enhance the accuracy and diagnostic efficiency of deep learning models on commonly used medical images.

Even though the selection of preprocessing methods and DL models is often influenced by factors such as the specific disease being diagnosed and the type of medical image used, there is no extensive study that systematically evaluates the impact of these techniques across a wide range of medical image datasets. Important variables, such as dataset size, the type and number of diseases to classify, the unique characteristics of medical images, and the specific features of each DL model, along with how preprocessing affects these factors, remain largely unexplored. Conducting such a study is crucial to providing

an objective evaluation and offering researchers valuable insights into selecting optimal methods for medical image classification.

Additionally, another area of improvement lies in the limited exploration of resource efficiency in the application of DL models for medical imaging. While model accuracy is frequently emphasized, the computational costs and resource demands – crucial for real-world clinical deployment – are often overlooked.

Moreover, there is a lack of practical guidance for professionals on selecting the most appropriate preprocessing techniques and DL models tailored to specific types of medical imaging datasets. This leaves practitioners with limited resources for optimizing their approaches based on the unique characteristics of their datasets, leading to suboptimal performance in both research and clinical settings.

***Our Work:***

Our study aims to fill these gaps by systematically exploring the impact of various preprocessing techniques when paired with advanced DL models applied to various types of medical imaging modalities. We aim to provide a comprehensive evaluation that not only enhances the understanding of how preprocessing can optimize diagnostic accuracy but also offers practical guidance for selecting the most effective preprocessing methods and DL models for specific types of medical imaging datasets. Additionally, we assess resource efficiency to ensure the feasibility of practical deployment in real-world settings.

To conduct our analysis, we applied these preprocessing methods alongside DL models to eight distinct datasets, each representing four common types of medical images. Specifically, we investigated two sets of X-ray images: the COVID19_Pneumonia_Normal_Chest_PA_Dataset [15] and the Osteoporosis Knee X-ray Dataset [16], two CT image collections: the Chest CT-Scan Images Dataset [17] and the Brain Stroke CT Image Dataset [18], two MRI datasets: Breast Cancer Patients MRI's [19] and Brain Tumor MRI Dataset [20], and finally, two ultrasound datasets: Ultrasound Breast Images for Breast Cancer [21] and MT_Small_Dataset [22].

Our research makes five significant contributions:

1. Comprehensive Evaluation of Preprocessing Techniques and DL Models: We systematically evaluate the impact of different preprocessing techniques (CLAHE + Butterworth, DWT + Threshold, CLAHE + Median Filter, Hybrid Median-Mean Filter, and Unsharp Masking + Bilateral Filter) paired with deep learning models (EfficientNet-B4, ResNet-50, DenseNet-169, VGG16, and MobileNetV2). This evaluation covers eight diverse datasets across various medical imaging modalities, offering a thorough analysis of how these combinations influence diagnostic performance.
2. Empirical Experimentation: The study includes extensive empirical experiments to confirm the impact and performance of the preprocessing techniques and deep learning models. By applying these methods and models to real-world medical imaging datasets, we provide evidence-based insights into their performance, addressing practical challenges such as noise reduction and feature extraction.
3. Insightful Analysis of Model-Preprocessing Combinations: We provide a thorough analysis and insights into which combinations of preprocessing techniques and deep learning models perform best for various types of medical images. This includes exploring how preprocessing methods can boost model performance and enhance diagnostic accuracy.
4. Guidance on Optimal Techniques for Medical Imaging: Based on our findings, we provide practical recommendations for selecting the best preprocessing methods and deep learning models for various medical imaging scenarios. This guidance aims to support researchers and practitioners in optimizing their approaches to medical image analysis.
5. Assessment of Resource Efficiency: In addition to evaluating effectiveness, we assess the usability and resource efficiency of the deep learning models. This includes an analysis of computational efficiency and runtime, helping identify models that achieve high performance while being resource-efficient.

These contributions advance the field by integrating algorithmic adaptation with experimental validation, aiming to enhance both the accuracy and efficiency of diagnostics in medical imaging. The remainder of the article is structured as follows: Section 2 discusses related work, Section 3 outlines the details of our proposed system, Section 4 analyzes the experimental results, and Section 5 summarizes our conclusions.

**Table 1**
The number of related literature by each publisher when searching by DL model.

| Publisher with search criteria | The number of literatures | | |
|---|---|---|---|
| | After downloading | After reviewing title and abstract | After reviewing full text |
| **Springer** | 1,294 | 45 | **6** |
| **IEEE** | 961 | 57 | **22** |
| **Elsevier** | 1,103 | 24 | **4** |
| **Nature** | 890 | 19 | **2** |
| **Total** | 4,248 | 145 | **34** |
| Others with unlimited publisher and pub. Year | | | **4** |
| **Final Review** | | | **38** |

## 2. Related works

This section reviews recent studies on DL models and preprocessing techniques for medical image classification, guided by credible research [23,24] and adhering to the PRISMA 2020 guidelines (Fig. 3) [25]. The criteria and methodology for selecting and evaluating relevant articles are described in Fig. 2.

### 2.1. DL models in medical image classification

After conducting a search for articles on DL models and medical images that fit the defined criteria, using the advanced search function in the library, a total of 4248 articles were collected. Following a review of the article titles and abstracts, and focusing on those ranked as Q1 and Q2, 145 articles were deemed relevant. Through a thorough content screening, 34 articles were selected as the most relevant to our study. Through meticulous examination of their content, 34 articles were selected as most pertinent to our research. In order to ensure comprehensive coverage and prevent overlooking articles from various sources, we extended our search to Google Scholar using similar keywords but without constraints on publishers or publication dates. This yielded an additional four pertinent studies, bringing the total of related articles to 38. The breakdown of articles per publisher can be found in Table 1, there are 6 studies related to EfficientNet-B4 [26–31], 6 studies on ResNet-50 [32–37], 7 studies on MobileNetV2 [38–44], 6 studies on VGG16 [45–50], and 8 studies on Densenet169 [51–58] and 5 studies on experiments or surveys [1,4,59–61]. Subsequently, we proceed with the content analysis of these 38 articles.

Recent advancements in the application of DL methods to classify diseases through medical images have been substantial and promising. DL has become fundamental in medical imaging diagnosis, renowned for its remarkable speed and accuracy [1,4,59]. The field's evolution is propelled by novel architectures and training techniques, with a shift toward using pre-trained models that utilize transfer learning. These models, including ResNet, VGG, DenseNet, MobileNet, and EfficientNets, are proving to be highly effective through the use of transfer learning, achieving accuracy rates frequently surpassing 90% in various studies [60,61].

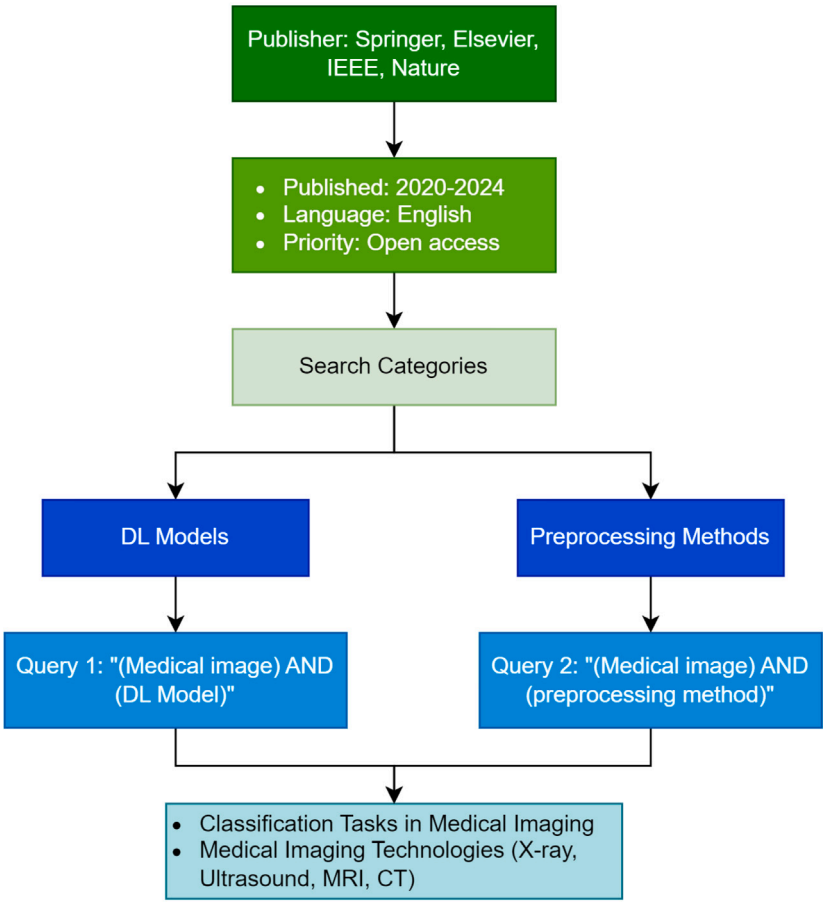



**Fig. 2.** The methodology for selecting and evaluating relevant articles.

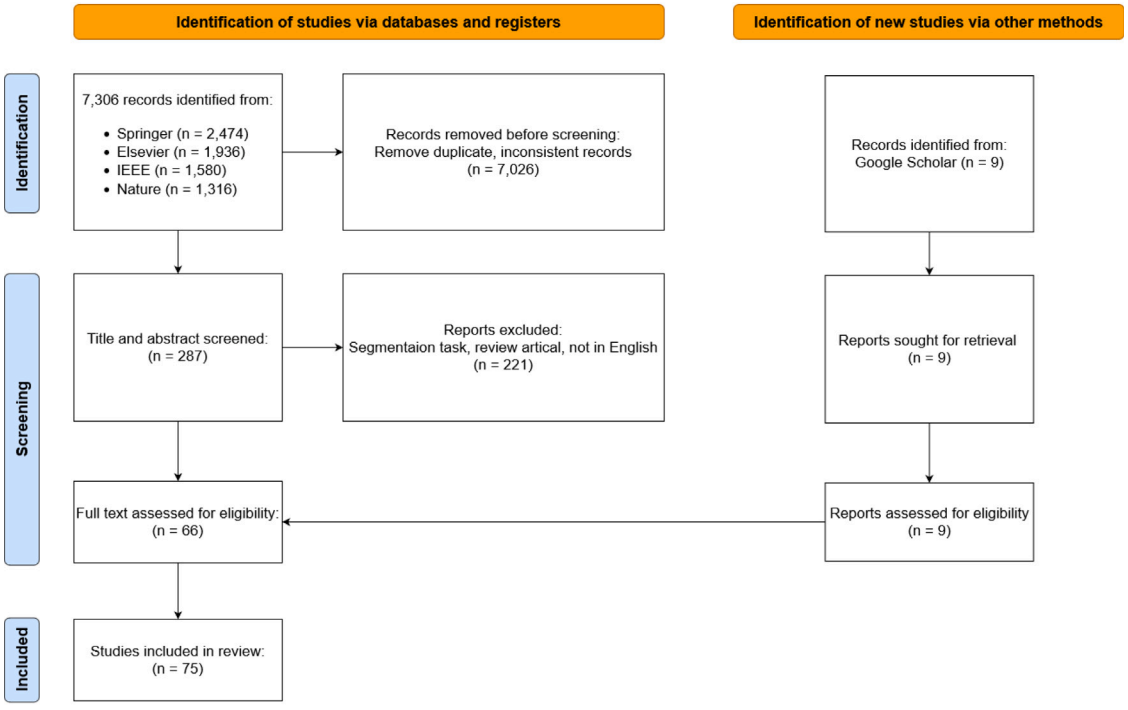



**Fig. 3.** PRISMA 2020 flow diagram.

EfficientNet-B4 has demonstrated high effectiveness across various medical imaging tasks, such as image segmentation, disease classification, and tumor detection. Specific applications include pulmonary edema and pleural effusion detection [27], identifying orthopedic hardware in pelvic and hip radiographs [26], diagnosing COVID-19 from CT and X-ray images [28,29], and brain tumor prediction using MRI images [30]. Its optimal balance between complexity and computational efficiency has made it a favored option among researchers. Moreover, EfficientNet-B4's adaptability enables smooth integration with various datasets and tasks, further highlighting its versatility and effectiveness in medical imaging applications [31].

Similarly, ResNet-50, a widely used CNN model, has been applied in various medical imaging tasks, such as MRI, CT, X-ray, and ultrasound analysis. Research has shown its effectiveness in applications such as diabetic retinopathy (DR) prediction [32], Classification of COVID-19 using chest X-ray (CXR) images [33], and accurate diagnosis of diseases like brain tumors through MRI scans [34,36]. ResNet-50 has also proven highly effective for lung disease classification using CT images [37] and breast cancer detection [35]. Its ability to extract discriminative features, leverage context sensitivity, and generate accurate predictions has been emphasized in these studies, demonstrating the model's strong classification capabilities across a diverse range of medical imaging modalities.

MobileNetV2 is extensively utilized in medical imaging tasks such as brain MRI analysis [42], detecting COVID-19 from CT and X-ray images [38,39], lung disease detection via chest X-ray [43], and breast cancer classification using ultrasound images [44]. Its lightweight architecture reduces computational costs and memory requirements, making it suitable for deployment on portable devices [40]. Researchers have optimized MobileNetV2 for efficient inference on resource-constrained embedded devices, achieving high verification accuracy rates and low power consumption [41]. The model's deep learning capabilities enable accurate recognition of various lung disorders, including COVID-19, bacterial, and viral pneumonia, highlighting its versatility in medical imaging applications. MobileNetV2's ability to balance accuracy and efficiency makes it a valuable tool for medical image analysis across various modalities.

The VGG16 model has been extensively studied in various medical imaging applications. It has delivered promising outcomes in areas like brain tumor detection with MRI scans [46], Lumbar Spinal Stenosis (LSS) diagnosis through DL with CT images [47], and identifying kidney stones in KUB X-ray images [48]. The model's effectiveness in accurately identifying different medical conditions from various imaging modalities underscores its versatility and potential for applications in MRI, CT [49], X-ray [45], and ultrasound image analysis [50].

DenseNet-169, a convolutional neural network architecture developed by Huang et al. [51], has demonstrated considerable potential in numerous medical imaging applications. In medical imaging, DenseNet-169 has been applied to tasks like fracture classification [52], prediction of renal anomalies for prenatal ultrasound diagnosis [53], functioning as a multilabel classifier for Alzheimer's disease based on MRI scans [54], and diagnosing COVID-19 from chest X-ray images [55,58]. Additionally, DenseNet-169 has been applied in combination with image enhancement and transfer learning methods in medical imaging applications, including breast cancer classification using mammography Images [56] and identifying COVID-19 infection from CT scans [57]. This highlights the versatility of DenseNet-169 across various domains when combined with appropriate methodologies. DenseNet-169 has demonstrated its value in medical imaging, offering high performance in tasks ranging from disease identification to fracture classification. Its dense connectivity structure and deep architecture make it well-suited for extracting intricate features from medical images, contributing to accurate diagnoses and classifications.

### 2.2. Preprocessing methods in medical image processing

A similar search using the second keyword structure, focused on preprocessing methods in medical images, identified 3,058 articles. Reviewing the titles and abstracts and focusing on Q1 and Q2 ranked papers led to 142 relevant articles. A detailed review selected 32 papers. An additional search on Google Scholar, without publisher constraints, identified 5 more relevant articles, totaling 37. Table 2 provides details of the articles categorized by publishers.

Wavelet thresholding is an effective method applied across multiple domains, including medical image denoising. This method involves suppressing noise by thresholding wavelet coefficients [62]. In the medical field, wavelet thresholding has been successfully applied to denoise ultrasound images, including minimizing speckle noise in medical ultrasound imaging [63]. Wavelet thresholding is highly effective due to its capacity to adjust to unknown data smoothness, which makes it a powerful method for image denoising [64]. This flexibility is vital in medical imaging, where minimizing noise is key to ensuring precise diagnoses and thorough analysis. Furthermore, wavelet coefficient thresholding is widely used for image denoising due to its simplicity and effectiveness [65]. Researchers have explored different thresholding strategies, such as hard and soft thresholding, to determine the optimal approach for denoising medical images [66]. Wavelet thresholding has been widely researched for image denoising and is recognized for its simplicity and effectiveness [67]. Moreover, wavelet thresholding has been applied in various medical imaging modalities, where it has shown promising results in noise reduction [65,66,68,69].

CLAHE is a widely recognized image enhancement method, recognized for its effectiveness in medical imaging applications [70]. The algorithm works on small image regions, enabling localized equalization. This feature is advantageous in medical imaging, as distinct areas may need varying enhancement levels [71]. CLAHE has shown superior effectiveness compared to other enhancement methods, supported by numerous studies [72].

Combining CLAHE with other preprocessing methods such as Butterworth filtering, Wiener filtering [73] or Median filtering [74] can provide further enhancements in medical image processing. The Butterworth filter, a commonly used low-pass filter in medical imaging, is utilized across multiple modalities, including MRI [75], ultrasound [76], X-ray [77], and CT [78] to improve image quality and aid in diagnostic procedures.

In medical image processing, the Butterworth low-pass filter and other techniques such as unsharp masking or CLAHE [77] are often applied as a final step [79]. This filter helps adjust contrast and smooth out the brightness gradient associated with tissue thickness at different locations on the image. The combination of CLAHE and Butterworth bandpass filter has been extensively applied in medical imaging, especially in DL tasks such as COVID-19 diagnosis and pneumonia detection [80].

Median filtering is a crucial technique in medical image processing, known for its versatility in smoothing, denoising, and removing artifacts in different imaging modalities. It is particularly effective in reducing noise while maintaining edges, making it valuable for medical image analysis [81]. Numerous studies have shown its superior effectiveness over other filters in eliminating salt and pepper noise in various medical images, including CT [81], ultrasound [82], MRI [83], and X-ray [84].

For improved image enhancement, incorporating median filtering with other preprocessing techniques, such as CLAHE, can be highly advantageous [85]. While CLAHE is adept at enhancing image contrast, it may introduce noise as a potential drawback [86]. Research indicates that applying median filtering following CLAHE effectively smoothens the foreground and eliminates the noise caused by CLAHE [87]. This amalgamated approach has shown successful results in processing Ultrasound Spinal Column Images [88], identifying COVID-19 in CXR images [89], and analyzing Skin Disease Images [85], underscoring its

versatility for broader applications in medical imaging.

Among the numerous filtering techniques, the mean filter is a widely used approach in medical image processing. Mean filters are widely used to denoise medical images affected by impulse noise. However, it is essential to note that their smoothing effect can come at the cost of blurring edges, which is often undesirable. The Median-Mean Hybrid filter offers a compelling alternative by employing the mean filter for initial smoothing followed by a targeted noise removal step with the median filter, ultimately preserving edges and achieving superior noise reduction [90,91]. Studies have shown the effectiveness of the Median-Mean Hybrid filter in medical image processing [92].

Unsharp masking and bilateral filtering are two effective techniques in medical image processing. Unsharp masking [93] enhances image sharpness but can introduce unwanted halos around edges, which requires careful adjustment [94]. Bilateral filtering [95] excels at preserving edges while reducing Gaussian noise, making it an effective edge-preserving and denoising filter [96]. A recent advancement in medical image denoising combines these two methods. This combined method efficiently eliminates noise while preserving sharp edges, showing considerable effectiveness in medical image processing [97,98].

## 3. Proposed methods

### 3.1. Overview method

Fig. 4 illustrates the overall experimental framework, detailing the integration of various preprocessing methods and DL models for medical image classification. Each dataset undergoes five different preprocessing methods, resulting in five preprocessed versions per dataset. These versions are then classified using five distinct DL models. We created 200 unique combinations by pairing five preprocessing methods with five DL models across eight datasets. The performance of these combinations was evaluated using five metrics: accuracy, precision, recall, F1-score and Kappa score. The results were then thoroughly synthesized, analyzed, and evaluated. To ensure consistency and reproducibility, all experiments are conducted within a singular computational environment, Google Colab, powered by the robust Tesla T4 GPU. The effectiveness of each model-dataset pair is evaluated based on two criteria: classification performance, measured by accuracy, precision, F1-score, recall, Kappa score, and resource efficiency, assessed using runtime. This approach provides a comprehensive assessment of classification performance.

### 3.2. Medical image preprocessing methods

Recent studies have demonstrated that combining medical image preprocessing techniques such as CLAHE, Butterworth filter, Median filter, Mean filter, Unsharp masking, and Bilateral filter can be more effective than using each method individually. Each preprocessing technique has its drawbacks; for example, Unsharp masking uses a linear high-pass filter, which is very sensitive to noise and can cause unwanted distortions in homogeneous areas of the image [98]. Adding the Bilateral filter helps to reduce noise and improve image quality [98]. Similarly, images enhanced by CLAHE may exhibit speckle noise [88], and the application of filters such as the Median filter [88] or Butterworth filter [80] afterward has been shown to further improve the images. The Mean filter can reduce noise but often blurs the output image [92], while the Median filter effectively removes noise but can also degrade edges [92]. Combining these two filters takes advantage of their strengths, enhancing noise reduction while preserving image details [92]. This underscores the significance of choosing and integrating suitable methods to attain the best possible outcomes in medical image processing.

#### 3.2.1. CLAHE and Butterworth

In the field of medical imaging, preprocessing is a crucial step to enhance image quality and facilitate better analysis. Two effective techniques used in tandem for this purpose are CLAHE and Butterworth Bandpass Filtering.

CLAHE is a more sophisticated technique than traditional histogram equalization, designed to enhance image contrast. Unlike the standard method that processes the entire image, CLAHE divides the image into small regions (tiles) and applies histogram equalization individually to each tile. This method prevents over-amplification of noise, a common problem with global histogram equalization, and enhances the local contrast more effectively. The parameters *clipLimit* and *tileGridSize* control the contrast limiting and the size of the contextual regions, respectively.

Butterworth Bandpass Filtering is a frequency domain filtering technique used to isolate specific frequency components from an image. It is particularly useful in medical imaging to remove noise and preserve essential structures. The Butterworth filter is characterized by its smooth transition between passed and attenuated frequencies, controlled by parameters like the lower and upper cutoff frequencies ($d_0$ and $d_1$) and the filter order (n). This smooth transition minimizes artifacts and ensures that essential features are retained while noise is suppressed.

The integration of CLAHE and Butterworth Bandpass Filtering in medical image preprocessing can be implemented using the following Algorithm 1. In this algorithm, the input image is initially transformed into a grayscale image using *cv2.cvtColor* from the OpenCV library, as color information is unnecessary for the preprocessing methods applied.

CLAHE is then applied to the grayscale image to enhance the local contrast. The *clipLimit* is set to 2.0, which restricts the contrast amplification to avoid noise enhancement, and *tileGridSize* is set to (8,8), which defines the size of the contextual regions for histogram equalization. After contrast enhancement, the image undergoes a transformation to the frequency domain via the Fast Fourier Transform (FFT), after which a Butterworth Bandpass Filter is applied. The parameters $d_0 = 15$ and $d_1 = 30$ define the lower and upper cutoff frequencies, while $n = 3$ specifies the order of the filter. These parameters are selected to optimize the balance between reducing noise and retaining image details. By integrating CLAHE and Butterworth filtering, this preprocessing method significantly improves the quality of medical images, ensuring that the subsequent analysis is more dependable and precise.

#### 3.2.2. DWT and threshold

The DWT is widely utilized for image and signal processing. Its effectiveness in image denoising stems from its capacity to localize signal energy into a limited number of coefficients across different scales. In the context of images, DWT serves to break down a 2D signal into various frequency channels that are spatially oriented and independent. These components, once decomposed, can be processed and then accurately reassembled into the original image using the Inverse Discrete Wavelet Transform (IDWT).

In medical image analysis, the *bior1.3* wavelet is critical for maintaining edge integrity, essential for precise diagnostics [99]. The symmetrical attributes of *bior1.3* enhance edge filtering [100], providing superior directional filtering and noise reduction capabilities, pivotal for isolating medical details from noise. *Bior1.3*'s balance between noise suppression and computational efficiency is critical in medical imaging, where swift processing is vital for real-time analysis or handling extensive datasets.

When applying the DWT to images, the image is decomposed into four frequency subbands through a process of filtering and downsampling. These subbands are labeled to as LL (Low-Low), LH (Low-High), HL (High-Low), and HH (High-High) [62], and each captures different aspects of the image's frequency content:

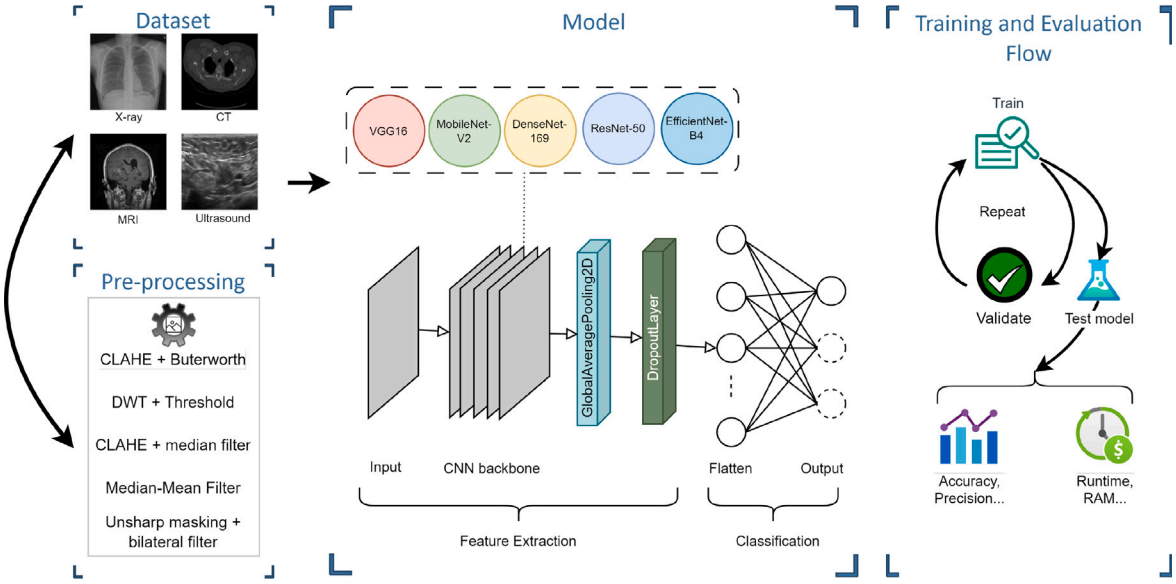



**Fig. 4.** The architecture of the proposed system.

**Algorithm 1:** Preprocess Image using CLAHE & Butterworth bandpass filter

**Description:** This algorithm enhances the contrast of images using CLAHE and filters frequencies using a Butterworth bandpass filter to improve image quality.

**Input:** $I_{in} \in \mathbb{R}^{H \times W \times C}$: input image
$d_0$: Lower cutoff frequency (default: 15)
$d_1$: Upper cutoff frequency (default: 30)
$n$: Filter order (default: 3)
**Output:** $I_{filtered} \in \mathbb{R}^{H \times W}$: Filtered image

$I_{gray} \leftarrow C_{gray}(I_{in})$ {Convert image to grayscale}
$I_{clahe} \leftarrow CLAHE_{clipLimit=2.0,tileGridSize=(8,8)} I_{gray}$ {Apply CLAHE}
$f \leftarrow D(I_{clahe})$ {Convert image to double precision}
$(n_x, n_y) \leftarrow \dim(I_{clahe})$ {Get image dimensions}
$fftI \leftarrow \mathcal{F}_{(2 \cdot n_x - 1, 2 \cdot n_y - 1)} f$ {Perform 2D FFT}
$fftI \leftarrow \mathcal{F}(f)$ {Shift FFT to center}
$butterworth_mask \leftarrow 0$ for $(2 \cdot n_x - 1, 2 \cdot n_y - 1)$ {Initialize Butterworth filter mask}
**for** $i := 0$ to $2 \cdot n_x - 1$ **do**
  **for** $j := 0$ to $2 \cdot n_y - 1$ **do**
    $dx \leftarrow i + 1 - (n_x + 1)$
    $dy \leftarrow j + 1 - (n_y + 1)$
    $dist \leftarrow \sqrt{dx^2 + dy^2}$
    $low_pass \leftarrow 1/(1 + (dist/d_1)^{2n})$ {Low-pass Butterworth filter}
    $high_pass \leftarrow 1 - 1/(1 + (dist/d_0)^{2n})$ {High-pass Butterworth filter}
    $butterworth_mask[i, j] \leftarrow low_pass \cdot high_pass$ {Band-pass filter}
  **end for**
**end for**
$fft_{filtered} \leftarrow fftI \cdot butterworth_mask$ {Apply Butterworth bandpass filter to FFT}
$I_{filtered} \leftarrow \mathcal{S}^{-1} fft_{filtered}$ {Shift FFT back}
$I_{filtered} \leftarrow \mathcal{F}^{-1}_{(2 \cdot n_x - 1, 2 \cdot n_y - 1)} I_{filtered}$ {Perform inverse 2D FFT}
$I_{filtered} \leftarrow \mathcal{R}_{[:n_x, :n_y]} I_{filtered}$ {Extract real part}
$\forall \text{pixel} \in I_{\text{filtered}}$, if pixel $< 0$ then pixel $= 0$, if pixel $> 255$ then pixel $= 255$, else pixel $=$ pixel.
$I_{filtered} \leftarrow \mathcal{U}(I_{filtered})$ {convert to 8-bit unsigned Integer}
**return** $I_{filtered}$

- LL: This subband represents the low-frequency components in both horizontal and vertical directions. It represents a smoothed approximation of the original image and contains the essential information, such as the overall structure and background. This subband is often called the approximation subband because it retains most of the image's significant features with reduced detail.
- LH: This subband holds low-frequency information in the horizontal direction and high-frequency information in the vertical direction. It primarily contains information about horizontal edges and variations within the image. This subband emphasizes the vertical transitions or changes in intensity.
- HL: This subband contains high-frequency information in the horizontal direction and low-frequency information in the vertical direction. It primarily contains information about vertical edges and variations within the image. This subband emphasizes the horizontal transitions or changes in intensity.
- HH: This subband contains high-frequency information in both horizontal and vertical directions. It contains information about fine textures, sharp edges, and noise. This subband highlights the detailed aspects of the image, such as fine lines, corners, and intricate textures.

**Algorithm 2:** Preprocess Image using DWT and Threshold

**Description:** This algorithm applies DWT for multi-resolution analysis of images and uses thresholding techniques to separate foreground from background, enhancing feature extraction.

**Input:** $I_{in} \in \mathbb{R}^{H\times W\times C}$: input image
**Output:** $I_{filtered} \in \mathbb{R}^{H\times W}$: Filtered image
$I_{gray} \leftarrow C_{gray}(I_{in})$
$coefficients \leftarrow DWT_{bior1.3}(I_{gray})$
$LL, LH, HL, HH \leftarrow coefficients$
$denoised_{HH} \leftarrow Thresh(HH, 0.5, \text{soft'})$
$denoised_{LH} \leftarrow Thresh(LH, 0.5, \text{soft'})$
$denoised_{HL} \leftarrow Thresh(HL, 0.5, \text{soft'})$
$N \leftarrow$ number of elements in $LL$
$\mu_{LL} \leftarrow \frac{1}{N}\sum_{i=1}^{N} LL_i$
$thresh_{LL} \leftarrow \sqrt{\frac{1}{N}\sum_{i=1}^{N}(LL_i - \mu_{LL})^2}$
$denoised_{LL} \leftarrow Thresh(LL, thresh_{LL})$
$coefficients \leftarrow (denoised_{LL}, denoised_{LH}, denoised_{HL}, denoised_{HH})$
$I_{filtered} \leftarrow DWT_{bior1.3}(coefficients)$
**return** $I_{filtered}$

Denoising typically follows three steps: wavelet decomposition, thresholding noise-related coefficients, and reconstruction. In Algorithm 2, the input image is converted to grayscale and subjected to multi-resolution analysis using the biorthogonal wavelet transform (*bior1.3*). The high-frequency subbands (LH, HL, HH) are denoised using soft thresholding with a fixed threshold value of 0.5 to suppress noise while preserving significant features. An adaptive threshold is computed for the low-frequency subband (LL) using the standard deviation of its coefficients, allowing for more efficient noise reduction. Lastly, the denoised subbands are merged through the inverse DWT to rebuild the filtered image, improving it by minimizing noise while highlighting key features.

Both hard and soft thresholding are essential for reducing noise and preserving vital image details in the realm of DWT for image denoising. The threshold value selected can greatly impact the denoising outcomes.

**Thresholding**

Hard thresholding is an effective method for removing intense noise by completely discarding components with values lower than a specified threshold. However, this approach can create images with abrupt transitions due to the drastic removal of low-value components. In contrast, soft thresholding preserves small and vital details better than hard thresholding. Although less effective in noise reduction compared to hard thresholding, this method is suitable for images with low to moderate noise levels. Fig. 5 illustrates the differences between hard and soft thresholding.

Adaptive thresholding automatically adjusts the threshold based on the characteristics of each image, balancing noise removal and detail preservation. It is particularly effective for complex images or subbands like LL that contain crucial details. However, this method requires additional computations to determine the adaptive threshold, making it more complicated than fixed thresholding methods.

Hard thresholding is commonly used in high-contrast images, such as X-ray images, while soft thresholding is more suitable for images with intricate details that need to be preserved, such as MRI brain scans. Adaptive thresholding is widely employed across various medical images [10], including ultrasound and complex medical imagery, due to its ability to adapt to specific image characteristics effectively.

The selection of an appropriate threshold is crucial, especially for the LL subband, also known as the low-resolution approximate image, which contains many intricate details. If the threshold is too high, minor yet significant details might be lost, leading to blurring and removing crucial features like edges, corners, or smooth transitions. On the other hand, if the threshold is set too low. In that case, it may not effectively remove noise, leaving residual noise in the final image, which reduces the effectiveness of the denoising process and negatively impacts the overall signal quality. Therefore, to efficiently denoise without losing important details in this subband, a threshold based on the standard deviation concept is applied [10].

### 3.2.3. CLAHE and median filter

The combination of CLAHE and median filtering is another powerful preprocessing method in medical imaging. This method aims to enhance image contrast while effectively reducing noise. CLAHE improves the local contrast of the image, making fine details more visible, while the median filter smooths the image, reducing noise without blurring edges. The algorithm for applying CLAHE followed by a median filter is outlined below Algorithm 3:

**Algorithm 3:** CLAHE and Median Filter

**Description:** This algorithm integrates CLAHE for enhancing contrast with a Median Filter for noise reduction, with the goal of improving image clarity.

**Input:** $I_{in} \in \mathbb{R}^{H\times W\times C}$: input image
**Output:** $I_{filtered} \in \mathbb{R}^{H\times W}$: preprocessed grayscale image
$I_{gray} \leftarrow C_{gray}(I_{in})$
$I_{clahe} \leftarrow CLAHE_{clipLimit=2.0, tileGridSize=(8,8)}\, I_{gray}$
$I_{filtered} \leftarrow \mathcal{M}_{median}(I_{clahe}, 3)$
**return** $I_{filtered}$

In this algorithm, the input image *img* is first converted to a grayscale image using OpenCV's *cv2.cvtColor* function. This step simplifies the image data, focusing on intensity rather than color. CLAHE is then applied to the grayscale image to enhance local contrast. The parameters *clipLimit* and *tileGridSize* are assigned values of (8,8), respectively, to control the extent of contrast enhancement and the size of the local regions processed. Once the contrast has been enhanced, a median filter is applied to the image using *cv2.medianBlur* with a kernel size of 3. The median filter efficiently reduces noise by substituting each pixel's value with the median of its surrounding pixels. This is especially helpful in eliminating salt-and-pepper noise while maintaining edge details. The function returns the filtered image, which now has enhanced contrast and reduced noise, making it more suitable for subsequent analysis and interpretation in medical applications. By integrating CLAHE and median filtering, this preprocessing approach enhances the clarity and quality of medical images, ensuring that important features are more discernible and that noise is minimized, facilitating more accurate diagnostics and analyses.

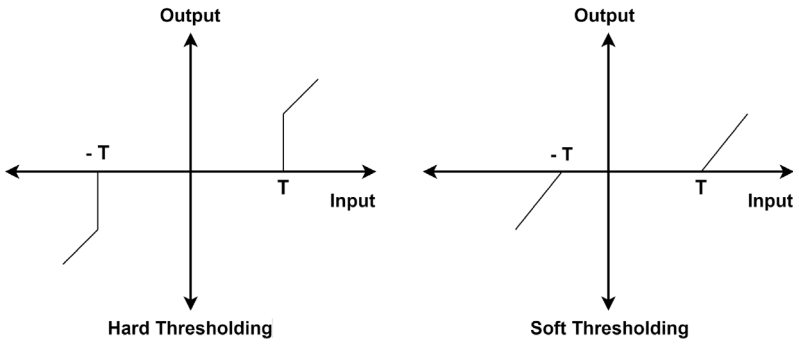



**Fig. 5.** Hard and soft thresholding.

*3.2.4. Median-Mean hybrid filter*

In image processing, filters are essential for noise reduction and detail preservation. Two commonly used techniques, the mean filter and the median filter, each have their strengths and limitations. The Median-Mean hybrid filter aims to combine the benefits of both to achieve superior noise reduction while retaining vital image details.

**Mean Filter:**

The mean filter, often referred to as the average filter, works by calculating the average pixel value within a defined neighborhood or kernel and replacing the center pixel with this computed average. This method smooths the image and is especially effective in minimizing Gaussian noise, thanks to its straightforward approach and low computational requirements. However, a significant drawback is its tendency to blur edges, leading to the loss of sharp details. Additionally, the mean filter is less against salt-and-pepper noise, which consists of sporadic high and low-intensity pixels.

**Median Filter:**

In contrast, the median filter substitutes each pixel's value with the median value of the surrounding pixels in its neighborhood. This is accomplished by arranging the pixel values in order and choosing the one in the center. For example, in a 3 × 3 kernel, the median is the 5th value after sorting. The median filter excels in removing salt-and-pepper noise without blurring edges, thereby preserving sharp details. However, it can distort delicate structures and textures, leading to a slightly unnatural appearance in some images.

**Median-Mean Hybrid Filter:**

**Algorithm 4:** Median Mean Hybrid Filter

**Description:** This algorithm utilizes a hybrid filtering method that merges the noise-reducing properties of the median filter with the smoothing effects of the mean filter, with the goal of retaining edges while minimizing noise in images.

**Input:** $I_{in} \in \mathbb{R}^{H\times W\times C}$: input image, $W_{size} \in \mathbb{Z}^2$: window size
**Output:** $I_{filtered} \in \mathbb{R}^{H\times W}$: Filtered image

$I_{gray} \leftarrow C_{gray}(I_{in})$
$p_h, p_w \leftarrow \frac{W_{size}[0]}{2}, \frac{W_{size}[1]}{2}$
$I_{padded} \leftarrow \text{Pad}_{p_h,p_w} I_{gray}$
$I_{output} \leftarrow \mathbf{0}_{H\times W}$ {Create ouput image zith zero values}
Loop through each pixel in the image with padding
**for** $x \leftarrow p_h$ **to** $H + p_h - 1$ **do**
  **for** $y \leftarrow p_w$ **to** $W + p_w - 1$ **do**
    $window_patch \leftarrow \text{ExtractSubwindow}(I_{padded}, x, y, p_h, p_w)$
    $V_{median} \leftarrow \text{median}(window_patch)$
    $V_{avg} \leftarrow \frac{1}{2}\left(\text{mean}(window_patch) + V_{median}\right)$
    $I_{filtered}[x - p_h, y - p_w] \leftarrow \frac{1}{|V_{avg}|}\sum_{i\in V_{avg}} i$
  **end for**
**end for**
**return** $I_{filtered}$

The hybrid filter integrates the advantages of both the mean and median filters, addressing the limitations of each. This approach effectively reduces various types of noise while maintaining crucial image details, making it especially useful for medical image preprocessing. Following Algorithm 4, the procedure starts by converting the color image to grayscale to facilitate easier processing. To handle edge pixels effectively, the image is padded with zeros (black). A predefined filter window size (often 3 × 3) is then used to traverse the image pixel by pixel. At each position of the filter window, a sub-window of surrounding pixels is captured. The median of these pixels is computed to address salt-and-pepper noise. Each pixel value within the sub-window is adjusted by adding the median value and dividing by two, thus smoothing the image while preserving crucial features. Ultimately, the center pixel in the output window is updated with the computed average of these modified values, ensuring that critical details are maintained while minimizing noise The image processed using the Median-Mean hybrid filter experiences both noise reduction and detail preservation, improving its overall quality and sharpness. This combined method ensures that the image maintains its key features, facilitating easier interpretation and analysis of the visual information. The combination of median and mean adjustments provides a balanced approach, minimizing the weaknesses of each individual filter and maximizing their strengths.

*3.2.5. Unsharp masking and bilateral filter*

**Unsharp Masking:**

Unsharp masking is a technique commonly used to sharpen image details. Despite its name, it does not involve actual blurring of the image. Instead, it focuses on highlighting edges and fine details.

To apply this technique, first, begin by applying a Gaussian filter to the original image to produce a softened version. Next, subtract the softened image from the original to obtain a detailed image (unsharp mask). Add this detailed image back to the original to boost edges and details effectively. Adjust the enhancement level using the scaling factor $\alpha$. The general formula is:

$$I_{\text{sharpened}} = I_{\text{original}} + \alpha \cdot (I_{\text{original}} - I_{\text{blurred}})$$

**Bilateral Filtering:**

A bilateral filter is a form of image processing filter utilized to decrease noise while retaining edges. It is a non-linear filter, with the degree of smoothing on a pixel dependent on the intensity variance with neighboring pixels and the spatial gap between them. These filters can smooth images by factoring in the spatial distance and intensity difference between pixels, preventing nearby pixels with differing intensities from being averaged together. Additionally, they are effective in reducing noise by averaging neighboring pixels' intensities to enhance image quality. Despite its benefits in noise reduction and edge preservation, this technique has drawbacks like high computational complexity, sensitivity to parameter configurations, and the risk of introducing artifacts. Additionally, it may struggle with various noise

**Table 2**
The count of relevant literature from each publisher when searching for preprocessing methods.

| Publisher with search criteria | The number of literatures | | |
|---|---|---|---|
| | After downloading | After reviewing title and abstract | After evaluating full text |
| **Springer** | 1,180 | 48 | **10** |
| **IEEE** | 975 | 69 | **19** |
| **Elsevier** | 477 | 14 | **2** |
| **Nature** | 426 | 11 | **1** |
| **Total** | 3,058 | 142 | **32** |
| **Others** sources with no restrictions on publisher and publication year | | | **5** |
| **Final Review** | | | **37** |

**Algorithm 5:** Unsharp Masking and Bilateral Filter

**Description:** This algorithm employs Unsharp Masking to enhance edges and details in images, coupled with a Bilateral Filter to smooth the image while preserving edges, optimizing both detail enhancement and noise reduction.

**Input:** $I_{in} \in \mathbb{R}^{H\times W\times C}$: input image
**Output:** $I_{filtered} \in \mathbb{R}^{H\times W}$: Filtered image

$I_{gray} \leftarrow C_{gray}(I_{in})$
$\text{blurred}(x,y) \leftarrow \sum_{i=-3}^{3}\sum_{j=-3}^{3} \frac{1}{2\pi\cdot 3^2}\exp\left(-\frac{i^2+j^2}{2\cdot 3^2}\right) I_{\text{gray}}(x+i,y+j)$
$\text{unsharp_mask}(x,y) \leftarrow I_{gray}(x,y) + 1.5\cdot\left(I_{gray}(x,y) - \text{blurred}(x,y)\right)$
Initialize $I_{filtered}$ as zero matrix of size $H\times W$
**for** $x = 1$ to $H$ **do**
  **for** $y = 1$ to $W$ **do**
    $G_{xy} \leftarrow I_{gray}(x,y) - \text{blurred}(x,y)$
    $W(x,y) \leftarrow \sum_{(i,j)\in N} g_{75}(||p-(i,j)||)\cdot g_{75}(I_{gray}(x,y) - I_{gray}(i,j))$
    $I_{filtered}(x,y) \leftarrow \frac{1}{W(x,y)}\sum_{(i,j)\in N} I_{gray}(i,j)\cdot g_{75}(||p-(i,j)||)\cdot g_{75}(G_{xy})$
  **end for**
**end for**
**return** $I_{filtered}$

types, including salt-and-pepper noise or structured noise.

**Unsharp Masking and Bilateral Filtering:**

Unsharp Masking enhances image details by emphasizing edges. When combined with Bilateral Filtering, these methods work together, enhancing overall image quality while maintaining crucial details. Bilateral Filtering ensures edge accuracy and reduces noise amplification that may occur with Unsharp Masking alone. This combined method greatly improves the clarity of medical images, revealing subtle details and enabling more accurate diagnoses and analyses. For more details, see Algorithm 5. The process begins by converting the input image to grayscale and applying a Gaussian filter to blur it, which decreases sharpness while enhancing the image's sensitivity. Then, Unsharp Masking sharpens the image by combining the original grayscale image and the blurred version with weights of 1.5 and −0.5, respectively, emphasizing edges and details. Subsequently, Bilateral Filtering smoothens the image while preserving edges, employing parameters like a weight function and a Gaussian function with a standard deviation of 75 and a specified range of influence. The result is a refined image with enhanced details and decreased noise, which is particularly beneficial for improving the analysis and diagnosis of medical images.

### 3.3. DL models

Despite their notable efficiency in image classification, CNNs require a large volume of training data, posing a significant challenge in sensitive fields like medicine with limited access to labeled datasets. To overcome this obstacle, transfer learning emerges as a powerful tool.

**Table 3**
Parameters of different models.

| Model | Total params | Trainable params | Non-trainable params |
|---|---|---|---|
| EfficientNet-B4 | 19 510 880 | 1 837 057 | 17673823 |
| ResNet-50 | 25 686 913 | 2 099 201 | 23587712 |
| DenseNet-169 | 14 348 865 | 1 705 985 | 12642880 |
| VGG16 | 15 241 025 | 526 337 | 14714688 |
| MobileNetV2 | 3 570 753 | 1 312 769 | 2257984 |

CNNs are adept at extracting image features, unveiling intricate patterns through each layer. Transfer learning takes advantage of pre-trained models built on large datasets like ImageNet and adapts them for tasks such as disease classification in medical images. This method has notably enhanced the accuracy of medical image classification in diverse scenarios.

Our research applies transfer learning using five different models. Initially, we load pre-trained CNN models, specifically those from the ImageNet dataset. Subsequently, we remove the top fully connected layers, the "heads" tailored for object classification in ImageNet, as they are less adaptable to new tasks. To retain the models' feature recognition proficiency, we "freeze" the remaining layers, ensuring they remain unaltered throughout the training phase.

Afterward, we adapt the models for medical image classification by incorporating additional layers [101] on top of the pre-trained structure. The first layer is *GlobalAveragePooling2D*, which efficiently reduces each feature map from the pre-trained network to a single value. This is then followed by a *Dropout* layer with a dropout rate of 0.4. *Dropout* randomly disables 40% of the neurons during training, a method used to reduce overfitting by randomly deactivating 40% of the neurons during training. Next, we add a *Dense* layer with 1024 units and a *ReLU* activation function, introducing non-linearity and enabling the network to capture complex patterns in the extracted features. Finally, we add a single-unit *Dense* layer with *sigmoid* activation for datasets that have two classes. The final layer is designed for binary classification, enabling the model to differentiate between two categories, such as healthy and diseased tissue. In case the datasets have more than two classes, the single-unit *Dense* is replaced by n-units *Dense* Layer with *softmax* activation (n: class numbers). This *Dense* layer, using *softmax* activation, produces a probability distribution across the classes, allowing the model to classify instances in multi-class scenarios effectively. For example, if the dataset has 3 classes (such as healthy tissue, inflamed tissue, and cancerous tissue), the final *Dense* layer will contain three units and apply *softmax* activation to determine which class the model predicts for each input sample. We visualize the model architecture as Fig. 6.

By integrating the pre-trained CNN models with these custom layers, we effectively build completely new neural networks. These new models leverage the pre-trained weights for feature extraction while incorporating the custom layers to address the specific medical image classification problem at hand. This integration not only boosts overall performance but also strengthens the model's capacity to generalize to new, unseen data. The selected models vary significantly in size and

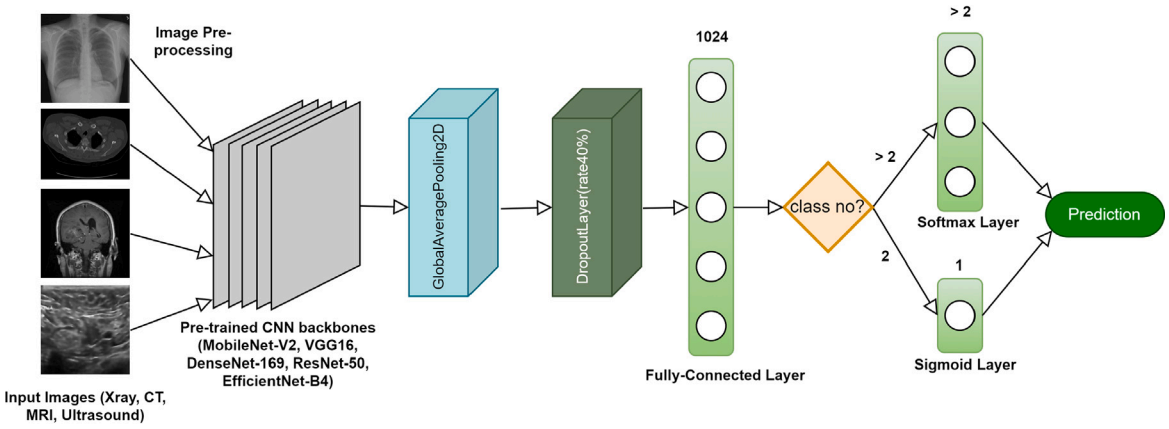



**Fig. 6.** Proposed model architecture.

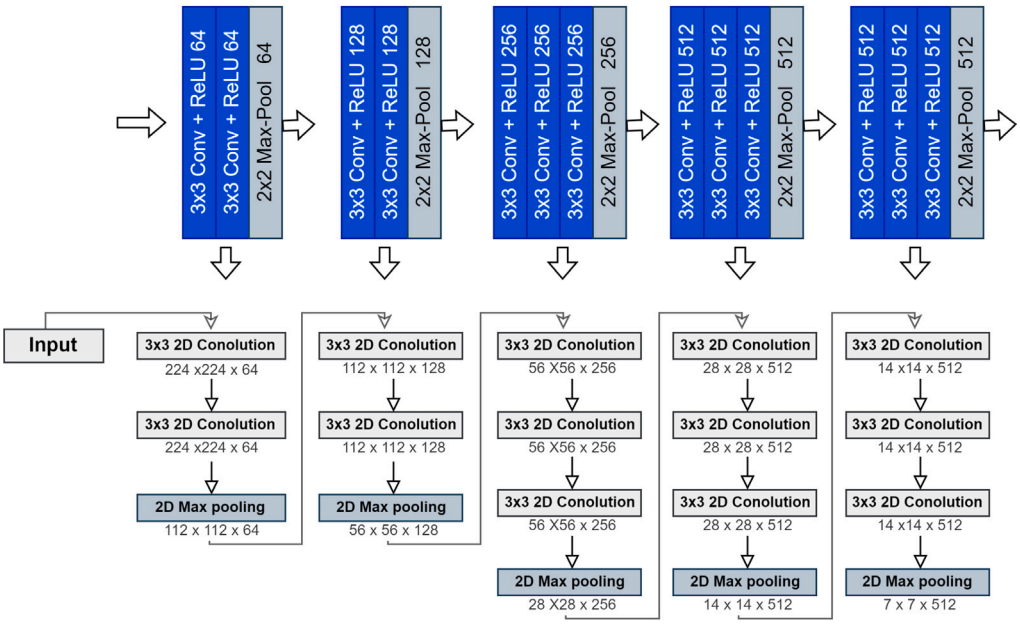



**Fig. 7.** The VGG16 backbone architecture.

complexity, ranging from lightweight architectures like MobileNetV2 to larger and more computationally intensive models like EfficientNet-B4. The details of each model, including the total number of parameters, trainable parameters, and non-trainable parameters, are summarized in Table 3.

### 3.3.1. VGG16

VGG16 [102], a convolutional neural network architecture created by the Visual Graphics Group at Oxford, has made a substantial contribution to the field of computer vision. Renowned for its simplicity and depth, VGG16 is composed of 16 weight layers, with 13 convolutional layers as shown in Fig. 7, and three fully connected layers. This architecture adopts small $3 \times 3$ convolutional filters. These layers are stacked to increase the network's depth, allowing it to capture intricate patterns in images. A key characteristic of VGG16 is the uniformity of its small filters, which enables the model to learn complex features while keeping computational demands reasonable.

The architecture's success can be attributed to its deep structure and the effective use of small convolutional filters, which allow for greater depth without a dramatic increase in computational complexity. VGG16 also incorporates max-pooling layers, which gradually reduce the spatial dimensions of the feature maps, helping to focus on key information and decrease the number of parameters. This method improves the network's capacity to generalize and excel in a range of tasks, including image classification, object detection, and transfer learning applications.

Despite the emergence of more efficient models, VGG16 continues to be a highly useful model in the field of computer vision. Its pre-trained weights, readily available online, offer a strong foundation for transfer learning. This enables researchers and developers to utilize VGG16's feature extraction abilities as a foundation for new tasks. By fine-tuning the pre-trained weights on a specific dataset, users can attain high accuracy in tasks such as object detection or image segmentation without having to train a large network from the ground up. This transfer learning strategy greatly decreases both training time and computational resource requirements, making VGG16 a flexible and accessible resource for various computer vision applications.

### 3.3.2. ResNet-50

ResNet-50 [103], or Residual Network-50, is a convolutional neural network architecture created by Microsoft Research in 2015. Presented in the groundbreaking study "Deep Residual Learning for Image Recognition", ResNet-50 has had a considerable impact on deep learning by tackling the vanishing gradient problem, which is a frequent challenge

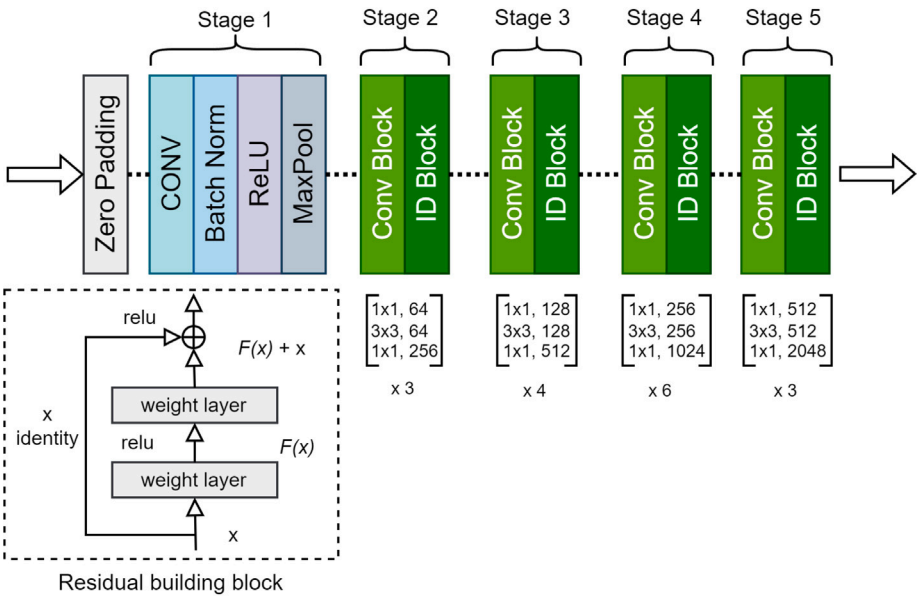



**Fig. 8.** The ResNet-50 backbone architecture.

encountered when training extremely deep neural networks. It is well-known for its ability to achieve high accuracy on image recognition tasks, particularly on the ImageNet dataset. ResNet-50 is a deep convolutional neural network with 50 layers that utilizes the residual network architecture. The backbone architecture of ResNet-50 (excluding the top layers) is shown in Fig. 8.

The residual network architecture is a breakthrough improvement over traditional CNN architectures. Instead of simply stacking convolutional layers, residual networks introduce residual blocks to help the network learn better. Each residual block contains two convolutional layers linked by a shortcut connection. This shortcut enables the input signal to be directly transmitted to the block's output, facilitating the network's ability to learn the identity function and mitigating the vanishing gradient issue often encountered in deep neural networks.

### 3.3.3. DenseNet-169

DenseNet-169 [51] is a variant of the DenseNet (Densely Connected Convolutional Networks) architecture, recognized for its efficiency and strong performance in medical image recognition applications. It has gained considerable attention in the realm of computer vision because of its capability to overcome the limitations found in conventional CNNs. Specifically, DenseNets effectively address the vanishing gradient problem, which occurs in deep neural networks when the gradients associated with earlier layers become very small. This difficulty hinders effective training. DenseNet-169 mitigates this issue by establishing connections between each layer and all subsequent layers, facilitating smoother information flow throughout the network and ensuring gradients remain substantial for training purposes. Furthermore, DenseNets bolster feature propagation. In traditional CNNs, each layer extracts features from the input image and then transmits them to the subsequent layer. However, valuable information can be lost during this process. DenseNet-169 combats this by connecting each layer to all subsequent layers. This enables more direct information flow from early to later layers, guaranteeing the network retains as much detail about the input image as possible. DenseNets also promote feature reuse. Traditional CNNs extract new features at each layer, which can be inefficient as some later-layer features might already be present in earlier ones. DenseNets circumvent this redundancy through connections between each layer and all subsequent layers. This allows the network to leverage previously extracted features, saving time and memory. Ultimately, DenseNets accomplishes a notable decrease in the number of parameters when compared to conventional CNNs. This makes them more efficient and easier to train, a quality particularly advantageous for deployment.

The structure of DenseNet-169 starts with a $7 \times 7$ convolutional layer using 64 filters, followed by a $3 \times 3$ max pooling layer. It comprises four Dense blocks with a structure as depicted in Fig. 9, separated by three transition layers, with the Dense blocks comprising varying numbers of layers: the first block has six layers, the second 12 layers, the third 32 layers, and the fourth 32 layers. Following the final Dense block, a global average pooling layer is included, which is then followed by a fully connected layer that uses a softmax activation function for classification purposes. DenseNet-169's architecture offers several advantages. It is parameter-efficient despite its depth, thanks to the Dense connections and feature map reuse. This architecture improves gradient flow, facilitating the training of deep networks. Additionally, it helps mitigate overfitting by reducing redundancy and encouraging feature reuse. DenseNet-169 is commonly employed in image classification, object detection, and segmentation tasks. Its efficiency and performance make it suitable for applications in medical imaging and real-time video analysis.

### 3.3.4. MobileNetV2

MobileNetV2 [104] is a CNN architecture tailored for mobile devices. It is an advancement of its predecessor, MobileNetV1, by further optimizing efficiency for mobile and embedded devices. Unlike VGG16, which prioritizes accuracy at the expense of computational efficiency, MobileNetV2 adopts a more lightweight approach. MobileNetV2 strikes a balance between performance and resource usage. This makes it ideal for applications where processing power and battery life are limited, such as smartphone cameras and on-device image recognition.

MobileNetV2 achieves its efficiency through the use of inverted residual blocks with linear bottlenecks as Fig. 10. These blocks contain a thin layer sandwiched between two wider layers. The network processes information primarily in this thin bottleneck layer, significantly reducing the computations required compared to traditional CNN architectures. Additionally, MobileNetV2 relies heavily on depthwise separable convolutions. This technique breaks down a standard convolution into two separate steps: a depthwise convolution that extracts features from individual channels of the input image, followed by a pointwise convolution that combines them into a single output channel. This separation significantly reduces computational cost without sacrificing the accuracy of the extracted features.

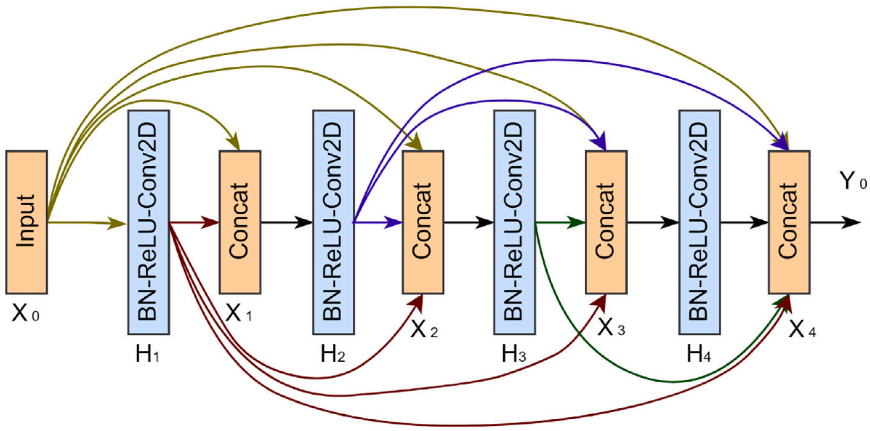



**Fig. 9.** A 5-layer Dense block in DenseNet-169 architecture.

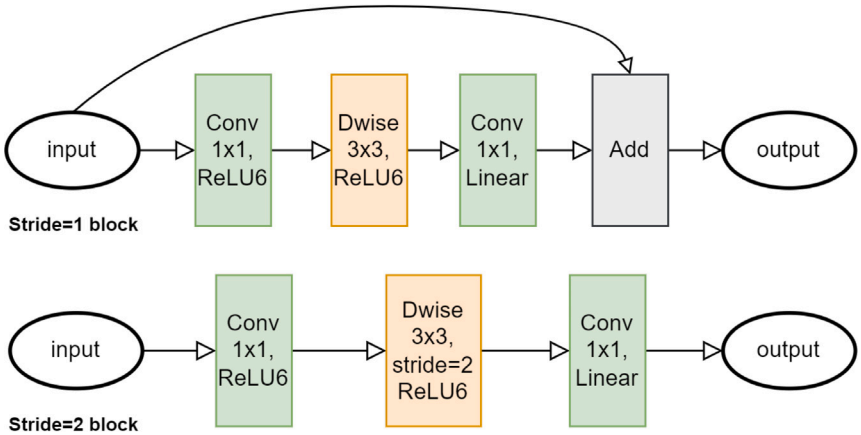



**Fig. 10.** Block types in MobileNetV2.

While MobileNetV2 prioritizes efficiency, it maintains high-performance levels. It delivers accuracy similar to more intricate models in a range of computer vision tasks, such as identifying images and detecting objects. This blend of efficiency and performance has established MobileNetV2 as a favored option for developers creating mobile apps needing real-time image processing features. The capacity to execute demanding tasks like object recognition on the device itself, without depending on cloud processing, presents notable benefits. This approach decreases latency, enhances privacy by retaining data on the device, and operates even without an internet connection — a revolutionary aspect for mobile applications.

However, MobileNetV2 has limitations. While it excels on mobile devices, its accuracy might not always match that of more powerful, compute-intensive models. Nevertheless, for tasks that require an optimal balance between efficiency and performance is crucial, MobileNetV2 remains a powerful and versatile tool in the computer vision toolbox.

### 3.3.5. *EfficientNet-B4*

EfficientNet-B4 [105] is a CNN model developed by Google specifically for image classification. It is part of a larger EfficientNet family known for achieving high accuracy while being computationally efficient.

One of the key innovations in EfficientNet is its compound scaling method. This method scales the network's depth, width, and resolution to maintain accuracy, unlike traditional approaches that focus on one dimension at a time. EfficientNet defines a set of coefficients that determine the network's size based on your needs. B4 represents a medium-sized model within the family. EfficientNet-B4's architecture has the primary component called MBConv (mobile inverted bottleneck) as Fig. 11.

SE blocks are incorporated to improve feature representation by dynamically adjusting the importance of each channel within the data. These blocks essentially pay more attention to informative features and suppress less relevant ones.

The overall structure of EfficientNet-B4 follows a basic inverted bottleneck design. The network is divided into sections: a stem, several stages, and a head. The stem preprocesses the input image, while stages are the building blocks that extract features. Each stage contains multiple blocks that share the same architecture but operate at different resolutions. These blocks are the workhorses of the network, and each one utilizes a compound operator, an SE block, and a transition layer. Transition layers adjust the resolution of the data as it flows through the network. Finally, the head transforms the features into the desired results, such as probabilities for different image categories.

Compound operators are another key component designed to lower computational demands while maintaining accuracy. They achieve this by combining two techniques: depthwise separable convolutions and linear bottlenecking. These convolutions decompose the standard convolution operation into two more efficient steps, while linear bottlenecking reduces the number of channels processed during the middle of the operation.

EfficientNet-B4 excels at classifying images into various categories, making it suitable for tasks like object recognition, scene understanding, and image content analysis. Its efficiency also makes it a strong candidate for transfer learning, in which a pre-trained model is adjusted for a particular task. This approach can significantly decrease the required time for training and enhance performance in comparison to

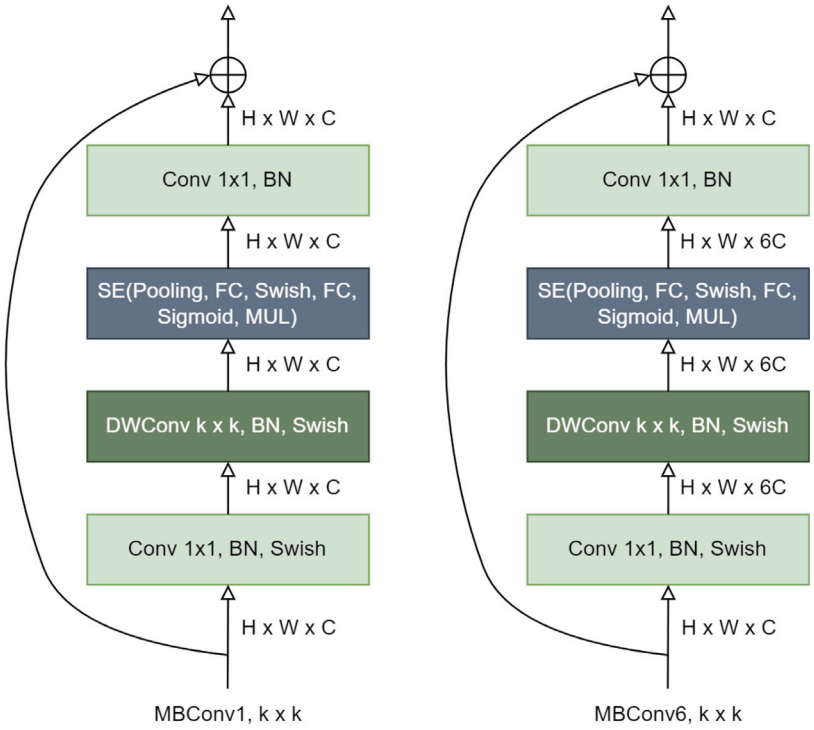



**Fig. 11.** MBConv components in EfficientNet-B4.

training a model from the ground up.

While EfficientNet-B4 provides an effective trade-off between efficiency and accuracy, even larger models within the EfficientNet family exist (B5-B7). These models can achieve even greater accuracy, but they require more computational requirements. For practical applications, particularly within medical diagnostic devices, resource limitations are a major concern. Therefore, we opted for EfficientNet-B4 as the optimal solution, prioritizing a balance between efficiency and resource consumption. This choice ensures the model can function effectively within the constraints of real-world medical devices.

## 4. Results - evaluation

We employ five types of evaluation metrics to evaluate the performance of various preprocessing combinations and DL models. We created 200 combinations by merging five preprocessing methods with 5 DL models across eight datasets. These 200 combinations were analyzed through five measures: accuracy, F1-score, precision, recall and Kappa score as shown in Fig. 12. The results were then synthesized, analyzed, and evaluated comprehensively.

### 4.1. Datasets

In this research, all used datasets are highly reliable and sourced from trusted repositories. A total of 8 datasets, covering multiple medical imaging techniques, including MRI, Ultrasound, X-ray, and CT, were collected for this study, as shown in Fig. 4, with variations in size across the original datasets. Some datasets are balanced, and some are not. For unbalanced datasets, although there is some related research that uses augmentation methods to balance the dataset, the team implements class weight in the training phase.

*a. Dataset 1 — Chest X-ray.* This dataset, publicly available on Kaggle [15], has been collected from various sources. It contains three folders corresponding to three classes (COVID-19, normal, pneumonia) in PNG, JPG, and JPEG formats and 2313 images per class. We divided dataset 1 into a training set and a testing set with ratio 80%:20% with random state 42 as Table 4, includes:

- The training set comprises 1880 samples belonging to Covid, 1837 images categorized as Normal, and 1835 images categorized as Pneumonia.
- The validation set consists of 433 images belonging to Covid, 476 images categorized as Normal, and 478 images categorized as Pneumonia.

This dataset supports various recent studies, including research on pre-processing methods for medical images [106] and the development of automatic diagnosis techniques [107].

*b. Dataset 2 — Knee X-ray.* it is publicly available on Kaggle [16] and comprises X-ray images of patients relating to osteoporosis knee, divided into two categories: Osteoporosis and Normal, with 186 samples in each group. In this study, for each class, by random state 42, we separated the dataset into two distinct sets as Table 4:

- Train set has 154 and 144 images for the Normal class and Osteoporosis class, respectively.
- Validation set has 32 and 42 images for the Normal class and Osteoporosis class, respectively.

This dataset contributes to several recent studies, including research on applying and improving CNNs to enhance the accuracy of knee osteoporosis diagnosis [108,109].

*c. Dataset 3 — Chest CT.* was published by Mohamed on Kaggle [17] and was collected from different sources. The dataset has 988 PNG images and 12 JPG images, which consist of 4 classes: 338 images of the Adenocarcinoma class, 187 of the Large cell class, 215 of the normal cell class, and 260 of the Squamous cell class. The dataset is already separated into three folders named train, valid, and test with distribution as Tables 4 and 5. We will keep this distribution ratio for the next steps.

This dataset contributes to several recent lung cancer detection studies, including research on applying the EfficientNet [110] model and improving the DenseNet model [111].

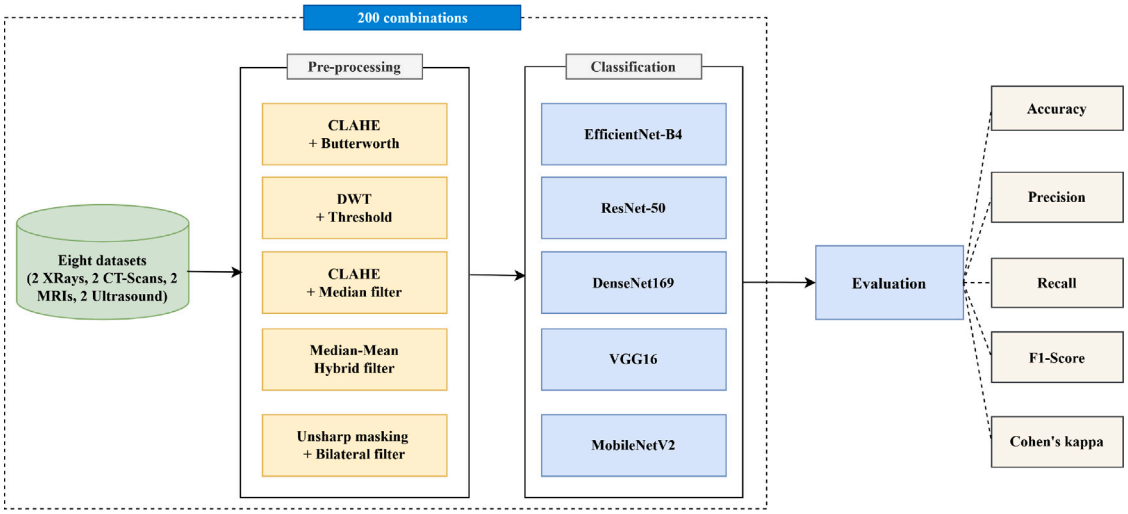



**Fig. 12.** Evaluation of classification performance on 200 combinations of preprocessing methods and DL models.

**Table 4**
Allocation of samples across datasets.

| Dataset | Short name | Full name | Total | Train | Val | Test |
|---|---|---|---|---|---|---|
| 1 | Chest X-ray | COVID-19 Pneumonia Normal Chest X-ray | 6939 | 5552 | – | 1387 |
| 2 | Knee X-ray | Osteoporosis Knee X-ray | 372 | 298 | – | 74 |
| 3 | Chest CT | Chest CT-Scan images | 1000 | 613 | 72 | 315 |
| 4 | Brain CT | Brain Stroke CT Image | 2501 | 2001 | – | 500 |
| 5 | Brain MRI | Brain Tumor MRI | 7023 | 5712 | – | 1311 |
| 6 | Breast MRI | Breast Cancer Patients MRI | 1480 | 1400 | – | 80 |
| 7 | Breast US.1 | Ultrasound Breast Images for Breast Cancer | 9016 | 8116 | – | 900 |
| 8 | Breast US.2 | MT Small | 400 | 320 | – | 80 |

**Table 5**
Dataset 3 — Chest CT distribution.

| | Adenocar-cinoma | Large cell | Squamous cell | normal cell |
|---|---|---|---|---|
| **Train** | 195 | 115 | 155 | 148 |
| **Val** | 23 | 21 | 15 | 13 |
| **Test** | 120 | 51 | 90 | 54 |

**Table 6**
Dataset 5 — Brain MRI distribution.

| | glioma | meningioma | no tumor | pituitary |
|---|---|---|---|---|
| **Train** | 1,321 | 1,339 | 1,595 | 1,457 |
| **Test** | 300 | 306 | 405 | 300 |

*d. Dataset 4 — Brain CT.* The dataset includes two classes, Normal and Stroke, with 1551 and 950 images, respectively, in JPG format [18]. In this study, as Table 4, the dataset 4 is split 80% for training and 20% for validation with random state 42:

- Train set has 1243 images labeled as Normal and 758 images labeled as Stroke.
- Validation set has 308 images labeled as Normal and 192 classified as Stroke.

This dataset has been used in studies to diagnose strokes, such as research on applying machine learning (ML) algorithms and developing predictive models for early detection [112], as well as research aimed at improving the performance of medical image classification [113].

*e. Dataset 5 — Brain MRI.* the collection includes JPG-format MRI scans of the human brain and comprises four classes — 1621 images of glioma class, 1645 of meningioma, 1757 of pituitary and 2000 of no tumor [20]. Dataset splits into two folders — Training and Testing as Table 4. Dataset distribution is as Table 6. We keep this dataset distribution for model input. The test set is also used for validation.

This dataset has been used to study brain tumor diagnosis using a newly developed model based on VGG19 [114], as well as to evaluate classification performance on ResNet152, VGG19, DenseNet169, and MobileNetv3 models [115].

*f. Dataset 6 - Breast MRI.* The dataset has breast cancer patients and healthy women images in JPG format [19]. There are two classes, Healthy and Sick, with a total of 740 images per class, which are already balanced. The dataset is split into two folders as Table 4:

- Training set includes 700 JPG images categorized as Healthy and 700 JPG images categorized as Sick.
- Testing set includes 40 JPG images categorized as Healthy and 40 JPG images categorized as Sick. We also use this test set as a validation set.

This dataset has been used to research medical image diagnosis using models such as MobileNet, MobileNetV2, NasNetMobile, VGG16, VGG19 [116], and DenseNet-201 [117].

*g. Dataset 7 - breast US.1:.* the dataset has ultrasound images [21] which are stored in 2 folders train and validation of women breasts under PNG and JPG formats as Table 4. There are also two classes — benign and malignant:

- The Train set is almost balanced with 4074 for benign and 4042 for malignant.
- The Test set has 500 and 400 images for benign and malignant, respectively. The test set is also used as a validation set.

This dataset is widely used to study the effectiveness of breast cancer classification using DL models such as VGG16, VGG19, MobileNetV2, and ResNet50 [118], as well as preprocessing methods like wavelet transformation to enhance CNN performance [119].

*i. Dataset 8 - Breast US.2.* The dataset has 1200 ultrasonic images of breast cancer patients with two classes, Benign and Malignant [22]. Each class has 200 original images, 200 Fuzzy-enhanced images, and 200 labeled images. In this study, original images are used and separated as Table 4 with random state 42:

- Train set has 163 and 157 images for Benign and Malignant, respectively.
- Test set has 37 and 43 images for Benign and Malignant, respectively.

This dataset was used in a prestigious study on breast cancer detection based on CNN [22,120]

### 4.2. Preprocessing

In this study, we implement five common preprocessing methods, which are:

1. **CLAHE with Butterworth:** CLAHE is applied to enhance image contrast, combined with a Butterworth filter for smoothing. CLAHE Parameters: *clipLimit* 2, *tileGridSize* 8 × 8; Butterworth parameters: lowcut 15, highcut 30, order 3.
2. **DWT with Threshold:** DWT is used for multi-resolution analysis, followed by thresholding to remove noise or enhance certain frequency components. Parameter: threshold 0.5.
3. **CLAHE + Median Filter:** CLAHE is applied to improve contrast, followed by a median filter that minimizes noise while retaining edges. This filter uses *ksize* 3.
4. **Median-Mean Hybrid Filter:** A hybrid filter combining median and mean filtering techniques designed to remove noise while preserving image details. Parameter: window size 3 × 3 (5 × 5 for dataset 2 - Knee X-ray)
5. **Unsharp Masking with Bilateral Filter:** Unsharp masking is employed for edge enhancement, coupled with a bilateral filter that softens the image while preserving edges and texture features. Parameters: *SigmaX* 3, *alpha* 1.5, filter size 9, *sigmaColor* 75 *sigmaSpace* 75.

Each sample is preprocessed with five methods. Most input images are in jpg and png format and are reprocessed by five methods. Output images are saved in jpg format. Although the jpg is a lossy compression format, the quality is 0.95 compared to the original image; the jpg still keeps high-quality image detail in a very small size. Each dataset is loaded by *image_dataset_from_directory*. *image_size* 240 pixel (380 pixel if Model is EfficientNet-B4), random seed 42, interpolation is bilinear, *batch_size* 32. If datasets are already split into train/val(test) separately, we will keep that ratio. If not, 80% of data will be used for training and 20% for validation.

### 4.3. Model configuration, training, and validation

In this research, we use five models (simple to complex): MobileNetV2, VGG16, DenseNet-169, ResNet-50, and EfficicientNet-B4. The backbone is kept as remain with weights is *imagenet*, default input_shape is 224 × 224 × 3 (380 × 380 × 3 if EfficientNet-B4), top is False. The backbone is Frozen with non-trainable parameters. The final layers will contain trainable parameters, including:

- **GlobalAveragePooling2D:** is used to convert the spatial dimensions of the input feature map to a vector by taking the average of all values.
- **Dropout with Rate 0.4:** Dropout regularization is applied using a rate of 0.4, which randomly deactivates a portion of input units throughout the training process to mitigate overfitting.
- **Dense 1024 with ReLU Activation:** A layer consisting of 1024 units and using the ReLU activation function is utilized to add non-linearity into the model.
- **Dense with Number of Outputs and Softmax/Sigmoid Activation:** Another fully connected layer with the appropriate number of output units is used, followed by either a softmax or sigmoid activation function. Softmax is typically used for scenarios involving multiple classes, while sigmoid is employed for binary classification problems.

Because most of the datasets are unbalanced, the team used the *compute_class_weight* technique to scope with this obstacle to help the training phase get better and more reliable results. The training will have 100 epochs; the learning rate is set at 0.001 and gradually reduces every five epochs by a factor of 0.9. If the training accuracy cannot get better within ten epochs, the training phase will be stopped.

For validation, a train-test split strategy is employed. If the datasets are not pre-split, an 80% training and 20% validation split is used to ensure a clear separation between training and evaluation phases. When datasets are pre-split, the existing split ratio is preserved to maintain consistency in evaluation.

During validation, the model's performance is assessed using accuracy, which measures the proportion of correctly classified instances out of the total number of instances. This metric provides a straightforward evaluation of how well the model generalizes to unseen data. Additionally, the validation process ensures that the best-performing model, in terms of the highest accuracy, is selected for final evaluation.

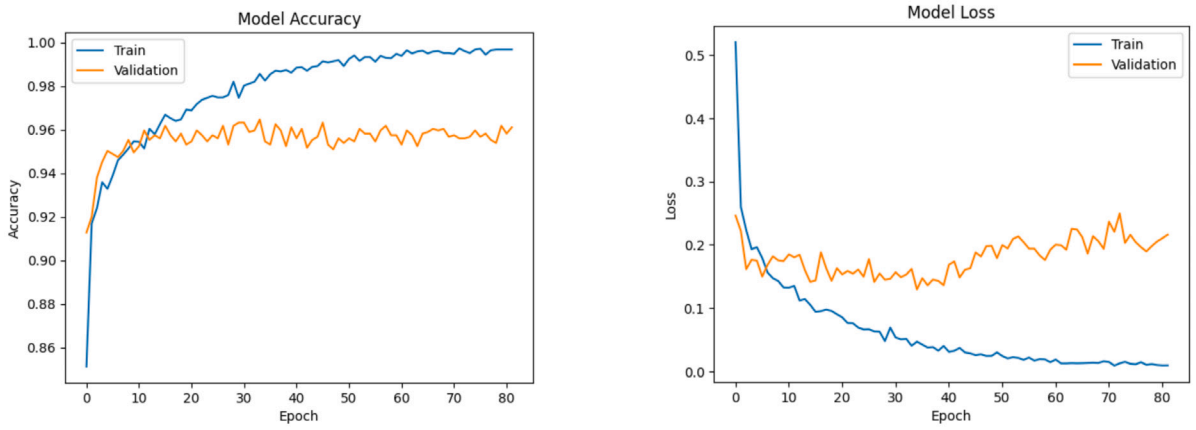
### 4.4. Evaluation metrics

In assessing the performance of ML model's performance, various metrics are employed to provide a comprehensive understanding of its effectiveness in solving the task at hand. The core metrics typically include Accuracy, Precision, Recall, and the F1 macro score. Core metrics:

- **Accuracy:** assess the ratio of correctly classified instances to the total number of instances. Although accuracy offers a clear indication of overall model performance, it may not be adequate when dealing with imbalanced datasets.
- **Precision:** Precision quantifies the proportion of true positive (TP) predictions among all positive predictions made by the model. It is calculated as the ratio of TPs to the sum of TPs and false positives (FPs). Precision is particularly important in scenarios where minimizing FPs is crucial.
- **Recall**, often referred to as sensitivity or the TP rate, evaluates the fraction of actual positive instances that are correctly recognized by the model. It is determined by the ratio of TPs to the total of TPs and false negatives (FNs). Recall is especially relevant in situations where detecting all positive instances is paramount.
- **F1 Macro Score:** represents the harmonic mean of recall and precision, offering a balanced assessment of both metrics. It is beneficial in multi-class classification problems where the class imbalance is present, as it computes the F1-score independently per class and then averages them.
- **Cohen's Kappa:** quantifies the level of agreement between the predicted and actual classifications, adjusting for what might occur by random chance. It accounts for both the observed agreement and the agreement expected by random chance, making it a robust metric for imbalanced datasets. Kappa values range from −1 (indicating complete disagreement) to 1 (indicating perfect agreement), while a value of 0 signifies no agreement beyond what would be expected by chance.

In addition to the core evaluation metrics, supplementary metrics related to computational resources and efficiency are also considered during model training. Additional Metrics, these include:

- **Average Run Time:** denotes the mean duration required for the model to analyze and make predictions on a given dataset. Monitoring average run time is essential for assessing the computational efficiency and real-world applicability of the model.




(a) Train and Val accuracy (b) Train and Val loss

**Fig. 13.** Accuracy and Loss in the training phase of Dataset 1 — Chest X-ray.

- **Random Access Memory (RAM) usage:** indicates the amount of system memory consumed by the model during training and inference. Managing memory usage is crucial for deploying models in resource-constrained environments.
- **GPU Utilization:** For models trained on GPU (Graphics Processing Unit) hardware, monitoring GPU utilization provides insights into how much GPU resources are utilized during training. Optimizing GPU utilization helps in maximizing training efficiency and reducing computational costs.

By considering both core evaluation metrics and additional metrics related to computational resources, practitioners can gain a thorough insight into the model's effectiveness and efficiency, thereby facilitating informed decision-making in model development and deployment.

### 4.5. Experimental results

#### 4.5.1. Performance metrics

Our experiment records 200 combinations (8 datasets, five preprocessing methods, and five models). To provide a concise summary, Table 7 presents the optimal combinations for each dataset, selecting the highest accuracy along with the four models that utilized the same preprocessing method. The evaluation metrics, including the same measures used in the training phase, Kappa score..., were computed using the test set. Besides, Fig. 19 presents pre-processing samples from 8 datasets, illustrating the best-performing images that achieved accurate predictions compared to other models utilizing the same preprocessing method.

**Dataset 1 - Chest X-ray:** Utilizing Dataset 1 - Chest X-ray [15] and employing a hydric preprocessing technique in conjunction with the ResNet-50 architecture. Accuracy and loss of training step are as Fig. 13, there is an early stop due to no-improvement backward epochs. We achieved a remarkable accuracy of 96.47%. The precision, recall, F1-score, and Kappa score were also commendable, with values of 96.75%, 96.56%, 96.55%, and 94.70%, respectively (see Table 7). Confusion Matrix (CM) is illustrated as Fig. 18a. Notably, this combination demonstrated promising results in accurately identifying COVID-19 cases by X-ray scans. Image 19(a) and 19(b) represent accurately predicted samples when employing ResNet-50, distinguishing them from the rest, which are incorrectly classified.

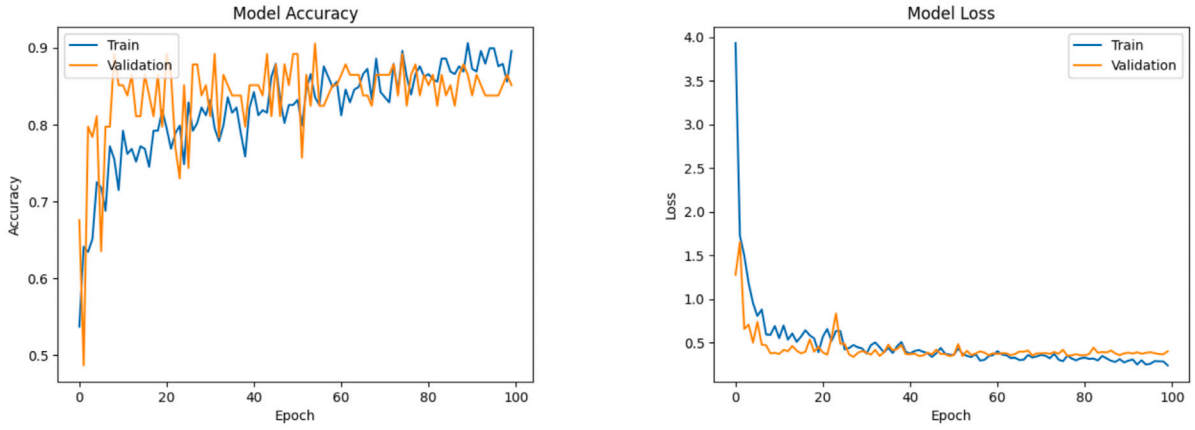
**Dataset 2 - Knee X-ray:** [16] underwent preprocessing using the unsharp masking and bilateral filter techniques, followed by training with the DenseNet-169 architecture. This approach yielded an impressive accuracy of 90.54%, with precision, recall, F1-score, and Kappa score values of 91.00%, 89.81%, 90.24%, and 80.51%, respectively (see Table 7). The training step is completed by 100 epochs with accuracy and loss measurements as Fig. 14. Fig. 18b presents a confusion matrix. The model demonstrated promising capabilities in accurately detecting signs of osteoporosis from X-ray images. When DenseNet-169 is utilized, Image 19(c) stands out as a correctly predicted sample, unlike the others, which are classified inaccurately.

**Dataset 3 - Chest CT:** Exploring Dataset 3 - Chest CT [17], we evaluated 25 combinations of preprocessing methods and models. Notably, the combination of DWT thresholding with the EfficientNet-B4 architecture achieved the highest accuracy of 87.62%, with precision, recall, F1-score, and Kappa score values of 88.19%, 89.59%, 88.75%, and 82.88%, respectively. CM is detailed in Fig. 18d. This indicates the efficacy of the selected preprocessing method in conjunction with a powerful model architecture in accurately diagnosing chest conditions from CT images. EfficientNet-B4 successfully predicts Image 19(d) as depicted, contrasting with the incorrect predictions for the other images.

**Dataset 4 - Brain CT:** Dataset 4 - Brain CT [18], preprocessed using the Median-Mean Hydric filter and trained with MobileNetV2, attained an outstanding accuracy of 96.20%. Moreover, the precision, recall, and F1-score metrics were equally impressive, highlighting the model's ability to properly detect signs of stroke from CT images. Fig. 18c presents the CM. The correct prediction of Image 19(e) is achieved with MobileNetV2, setting it apart from the incorrect predictions made for the other images.

Furthermore, we assessed the computational efficiency of the models by measuring their average run time, RAM consumption, and GPU utilization (see Fig. 15). As "Average run-time" chart, MobileNetV2 not only achieves the highest accuracy, but it also has the lowest resource consumption in terms of both accuracy and resource efficiency, with an average run time of just 6.97 s. To give a comprehensive comparison of the models' effectiveness, a Bubble chart is presented in Fig. 16. In this chart, the $x$-axis represents throughput (measured in images per second), the $y$-axis denotes accuracy (percentage), and the bubble size indicates the number of parameters in each model. This visualization highlights the trade-offs between speed, accuracy, and model complexity, illustrating how MobileNetV2 not only achieves the highest accuracy but also maintains a lower quantity of parameters compared to leftover models.

**Dataset 5 - Brain MRI:** For Dataset 5 - Brain MRI [20], employing the Median-Mean Hydric filter in conjunction with the EfficientNet-B4 resulted in an exceptional accuracy of 98.55%. The precision, recall, and F1-score provided additional confirmation of the model's proficiency in accurately detecting brain tumor disease by MRI scans. A confusion matrix is presented in Fig. 18e. Image 19(f) demonstrates a correct prediction when EfficientNet-B4 is applied, while the remaining images show incorrect predictions.




(a) Train and Val accuracy (b) Train and Val loss

**Fig. 14.** Accuracy and Loss in the training phase of Dataset 2 — Knee X-ray.

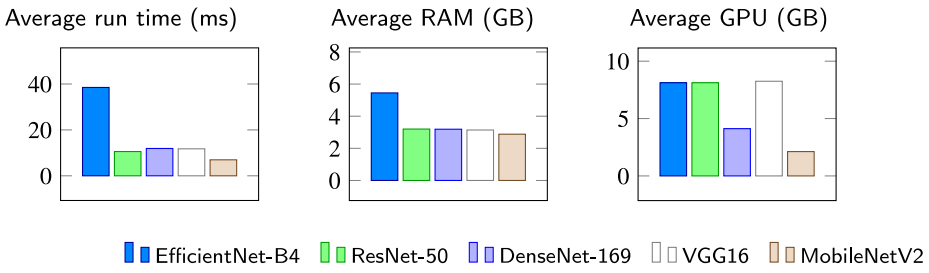



**Fig. 15.** Average run time, RAM, GPU usage between models in dataset 4 — Brain CT [18].

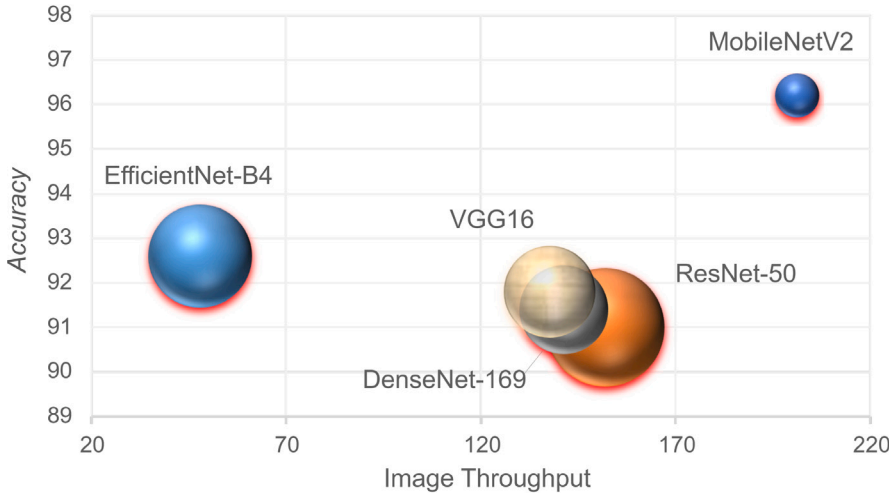



**Fig. 16.** Performance Overview in Dataset 4 — Brain CT.

**Dataset 6 - Breast MRI:** In the case of Dataset 6 - Breast MRI [19], preprocessing the images using unsharp masking and bilateral filtering led to an accuracy of 83.75% when trained with the MobileNetV2 model. Despite the slightly lower accuracy compared to other experiments, the other metrics values remained consistent, indicating the model's reliability in diagnosing breast conditions from MRI images. The kappa score was measured at 67.50%, further supporting the model's effectiveness. CM is presented in Fig. 18f. MobileNetV2 accurately identifies Image 19(g), distinguishing it from the images with incorrect predictions. The same with other datasets, among all models, MobileNetV2 stands out not only for its superior accuracy but also for its minimal resource consumption as Fig. 17.

**Dataset 7 - Breast US.1:** Combining unsharp masking and bilateral preprocessing with the EfficientNet-B4 architecture, the Ultrasound Breast Cancer dataset [21] achieved an impressive accuracy of 98.78%. The other metrics further reinforced the model's effectiveness in accurately identifying breast cancer from ultrasound images. Fig. 18g shows the Confusion matrix. Among the images, Image 19(h) is correctly predicted when EfficientNet-B4 is employed, contrasting with the incorrect predictions for the others.

Finally, **Dataset 8 - Breast US.2** [22] attained a remarkable accuracy of 96.25% when preprocessed using unsharp masking and bilateral filtering and trained with the EfficientNet-B4 model. The precision, recall, and F1-score metrics demonstrated the model's robustness in

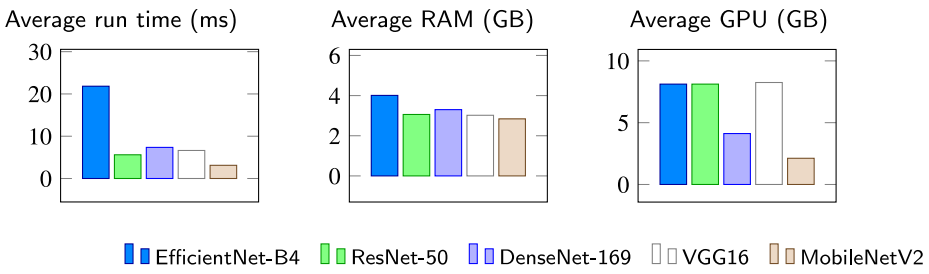



**Fig. 17.** Average run time, RAM, GPU usage between models in dataset 6 — Breast MRI [19].

**Table 7**
Evaluation metrics for eight datasets.

| Pre-processing | Model | Accuracy (%) | Precision (%) | Recall (%) | F1-score (%) | Kappa (%) |
|---|---|---|---|---|---|---|
| Dataset 1 - Chest X-ray | | | | | | |
| CLAHE + Butterworth | EfficientNet-B4 | 95.75 | 96.13 | 95.81 | 95.85 | 93.61 |
| DWT + Threshold | EfficientNet-B4 | 96.40 | 96.70 | 96.47 | 96.48 | 94.59 |
| CLAHE + median filter | EfficientNet-B4 | 95.96 | 96.26 | 96.03 | 96.06 | 93.94 |
| Median-Mean Hybrid Filter | ResNet-50 | **96.47** | 96.75 | 96.56 | 96.55 | 94.70 |
| Unsharp Masking + Bilateral Filter | EfficientNet-B4 | 96.32 | 96.57 | 96.41 | 96.40 | 94.48 |
| Dataset 2 - Knee X-ray | | | | | | |
| CLAHE + Butterworth | DenseNet-169 | 89.19 | 88.99 | 88.99 | 88.99 | 77.98 |
| DWT + Threshold | VGG16 | 86.49 | 86.24 | 86.24 | 86.24 | 72.47 |
| CLAHE + median filter | DenseNet-169 | 86.49 | 86.52 | 85.86 | 86.12 | 72.26 |
| Median-Mean Hybrid Filter | DenseNet-169 | 87.84 | 87.73 | 87.43 | 87.56 | 75.13 |
| Unsharp Masking + Bilateral Filter | DenseNet-169 | **90.54** | 91.00 | 89.81 | 90.24 | 80.51 |
| Dataset 3 - Chest CT | | | | | | |
| CLAHE + Butterworth | EfficientNet-B4 | 85.71 | 85.89 | 88.13 | 86.39 | 80.44 |
| DWT + Threshold | EfficientNet-B4 | **87.62** | 88.19 | 89.59 | 88.75 | 82.88 |
| CLAHE + median filter | EfficientNet-B4 | 84.76 | 85.71 | 88.56 | 85.81 | 79.31 |
| Median-Mean Hybrid Filter | VGG16 | 84.44 | 86.29 | 87.03 | 84.82 | 78.65 |
| Unsharp Masking + Bilateral Filter | EfficientNet-B4 | 86.03 | 86.50 | 89.39 | 86.94 | 80.97 |
| Dataset 4 - Brain CT | | | | | | |
| CLAHE + Butterworth | MobileNetV2 | 95.60 | 95.20 | 95.55 | 95.37 | 90.74 |
| DWT + Threshold | MobileNetV2 | 95.20 | 95.52 | 94.34 | 94.86 | 89.73 |
| CLAHE + median filter | EfficientNet-B4 | 92.60 | 92.65 | 91.64 | 92.09 | 84.19 |
| Median-Mean Hybrid Filter | MobileNetV2 | **96.20** | 96.12 | 95.84 | 95.97 | 91.94 |
| Unsharp Masking + Bilateral Filter | MobileNetV2 | 94.40 | 94.01 | 94.18 | 94.09 | 88.19 |
| Dataset 5 - Brain MRI | | | | | | |
| CLAHE + Butterworth | EfficientNet-B4 | 98.02 | 97.95 | 97.85 | 97.88 | 97.34 |
| DWT + Threshold | EfficientNet-B4 | 98.32 | 98.28 | 98.17 | 98.19 | 97.75 |
| CLAHE + median filter | EfficientNet-B4 | 97.94 | 97.89 | 97.76 | 97.77 | 97.24 |
| Median-Mean Hybrid Filter | EfficientNet-B4 | **98.55** | 98.52 | 98.42 | 98.43 | 98.06 |
| Unsharp Masking + Bilateral Filter | ResNet-50 | 98.47 | 98.41 | 98.34 | 98.36 | 97.95 |
| Dataset 6 - Breast MRI | | | | | | |
| CLAHE + Butterworth | MobileNetV2 | 76.25 | 79.35 | 76.25 | 75.61 | 52.50 |
| DWT + Threshold | MobileNetV2 | 81.25 | 82.24 | 81.25 | 81.11 | 62.50 |
| CLAHE + median filter | EfficientNet-B4 | 76.25 | 77.65 | 76.25 | 75.95 | 52.50 |
| Median-Mean Hybrid Filter | EfficientNet-B4 | 78.75 | 80.28 | 78.75 | 78.48 | 57.50 |
| Unsharp Masking + Bilateral Filter | MobileNetV2 | **83.75** | 83.77 | 83.75 | 83.75 | 67.50 |
| Dataset 7 - Breast US.1 | | | | | | |
| CLAHE + Butterworth | EfficientNet-B4 | 96.33 | 96.19 | 96.45 | 96.30 | 92.60 |
| DWT + Threshold | EfficientNet-B4 | 97.44 | 97.40 | 97.43 | 97.41 | 94.83 |
| CLAHE + median filter | EfficientNet-B4 | 96.44 | 96.42 | 96.38 | 96.40 | 92.80 |
| Median-Mean Hybrid Filter | EfficientNet-B4 | 97.44 | 97.35 | 97.50 | 97.42 | 94.83 |
| Unsharp Masking + Bilateral Filter | EfficientNet-B4 | **98.78** | 98.69 | 98.85 | 98.76 | 97.53 |
| Dataset 8 - Breast US.2 | | | | | | |
| CLAHE + Butterworth | EfficientNet-B4 | 90.00 | 90.16 | 89.75 | 89.90 | 79.81 |
| DWT + Threshold | EfficientNet-B4 | 90.00 | 89.94 | 89.94 | 89.94 | 79.89 |
| CLAHE + median filter | EfficientNet-B4 | 90.00 | 90.16 | 89.75 | 89.90 | 79.81 |
| Median-Mean Hybrid Filter | VGG16 | 91.25 | 91.25 | 91.48 | 91.24 | 82.50 |
| Unsharp Masking + Bilateral Filter | EfficientNet-B4 | **96.25** | 96.18 | 96.32 | 96.24 | 92.47 |

accurately diagnosing medical conditions from the provided dataset. A confusion matrix is presented in Fig. 18h. The prediction for Image 19(i) is accurate with EfficientNet-B4, whereas the predictions for the remaining images are incorrect.

#### 4.5.2. Metrics analysis

The pursuit of optimal performance in medical image processing and classification tasks is crucial for ensuring accurate diagnoses and improved patient outcomes. This section delves deeper into the evaluation metrics obtained from eight diverse datasets, exploring the

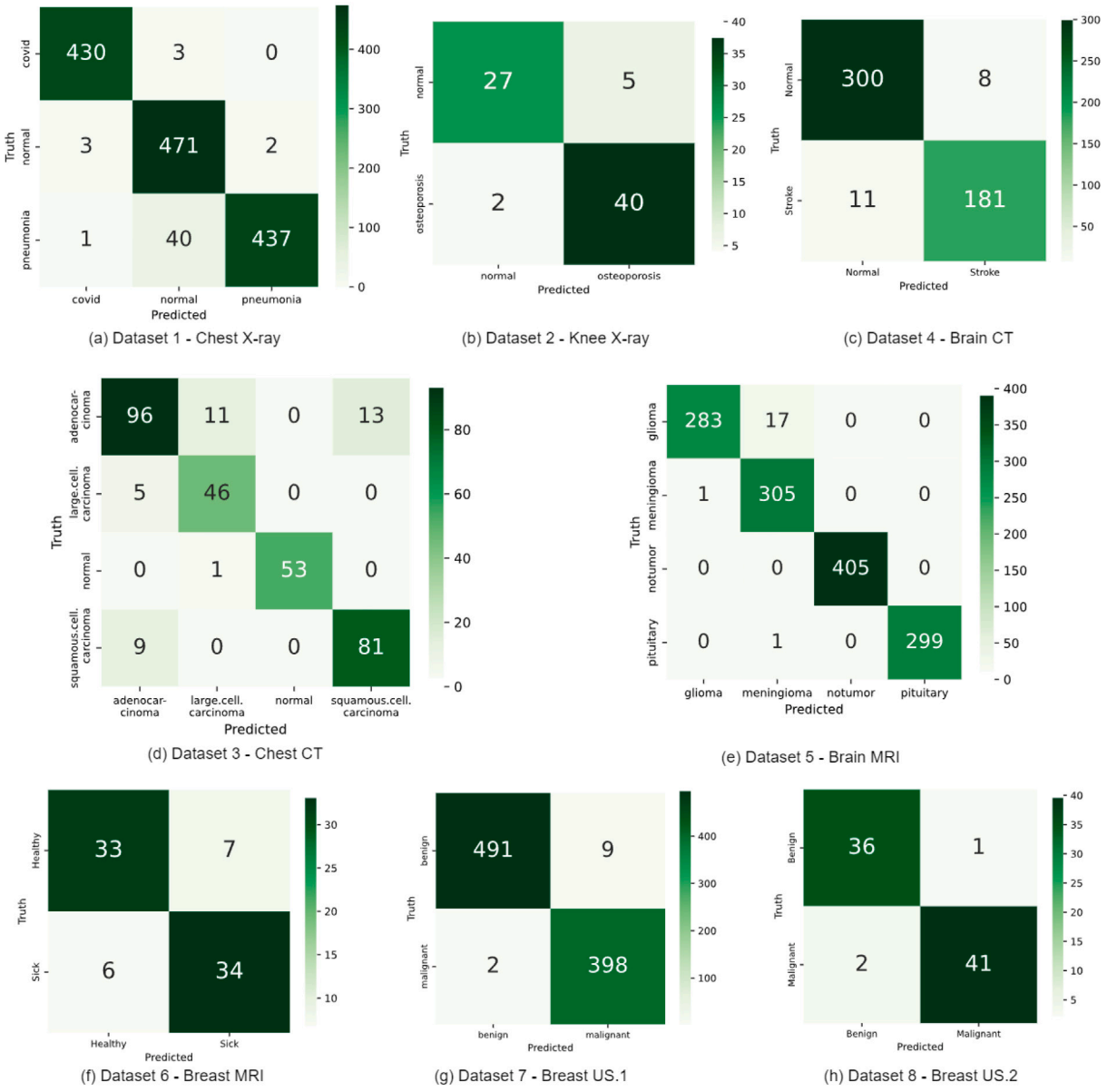



**Fig. 18.** Confusion matrix for 8 test sets.

most effective reprocessing methods and the models that yield the highest test accuracy for each dataset type as Table 7. Additionally, we examine the synergies between preprocessing techniques and model architectures to elucidate patterns of performance enhancement.

***Optimized Reprocessing Methods***

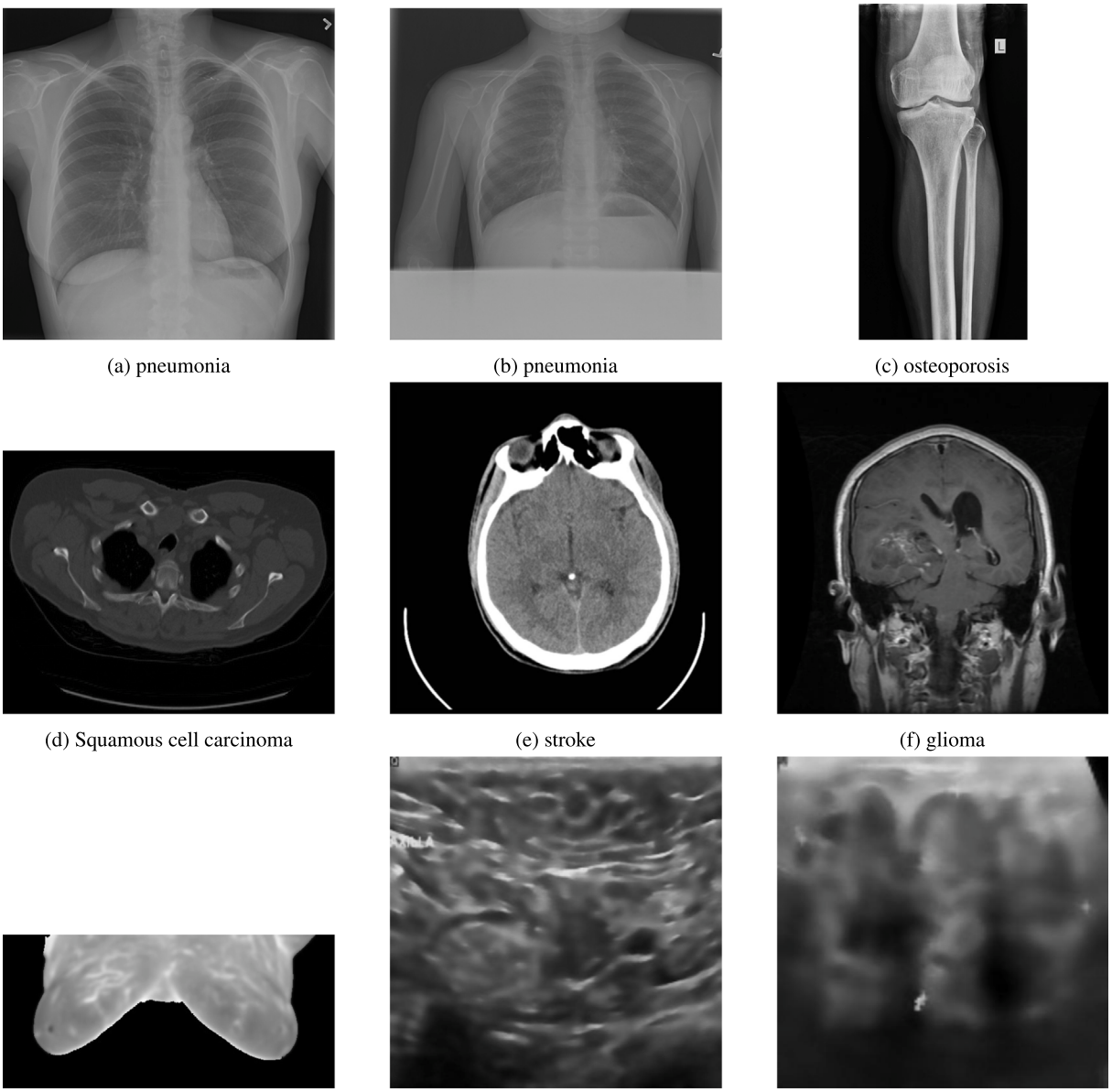
The analysis of reprocessing methods reveals insightful trends in improving model performance across various datasets as Fig. 20(b). Among the techniques employed, unsharp masking combined with bilateral filtering demonstrates promising results, particularly with a ratio of 4/8 (50%) for optimal performance. This method effectively enhances image sharpness while preserving important features through bilateral filtering, resulting in improved classification accuracy. Similarly, the median-mean hybrid filter, coupled with a ratio of 3/8 (37.5%), showcases significant enhancement in classification performance, underscoring the importance of hybrid approaches in balancing noise reduction and feature preservation. Overall, both the Median-Mean Hybrid Filter and the unsharp masking combined with a bilateral filter are highly effective preprocessing methods, achieving an 87.5% effectiveness ratio, corresponding to 7 out of 8 datasets.

***Optimized Models***

Our evaluation identifies EfficientNet-B4 and MobileNetV2 as the most effective models for different dataset types. Fig. 20(a) summarizes the number of datasets for which each model achieved optimal accuracy, represented in a bar chart. Each bar indicates the proportion of preprocessing methods that yielded optimal results, distinguished by different colors. EfficientNet-B4 exhibits superior performance across multiple datasets, accounting for 50% (4 out of 8) of the datasets. In these four datasets, it particularly excels in ultrasound image classification, where it achieves a ratio of 100% (2 out of 2 datasets). On the other hand, MobileNetV2 showcases its proficiency in handling diverse datasets, securing optimized accuracy in 25% (2 out of 8 datasets) of the cases. The versatility of MobileNetV2 underscores its adaptability to various imaging modalities, making it a compelling choice for medical image analysis tasks. Combined, EfficientNet-B4 and MobileNetV2 dominate the performance metrics, collectively covering 75% (6 out of 8) of the datasets.

***Synergistic Performance Enhancements***

An intriguing observation emerges when considering the combination of preprocessing techniques with model architectures. The synergy between EfficientNet-B4 and the unsharp masking-bilateral filter

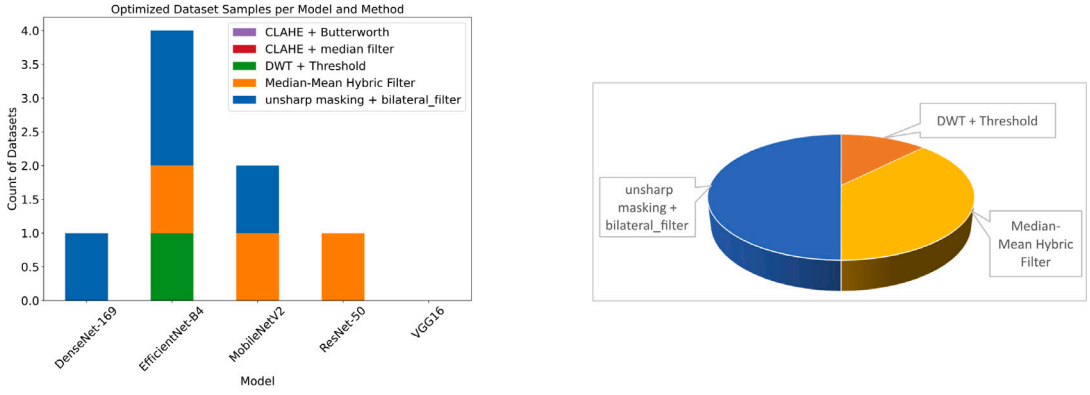


(a) pneumonia (b) pneumonia (c) osteoporosis

(d) Squamous cell carcinoma (e) stroke (f) glioma

(g) healthy (h) malignant (i) benign

**Fig. 19.** Pre-processing samples from 8 datasets. A, b, e, f: applied Median-Mean Hydric Filter; d: applied DWT + Threshold; c, g, h, i: applied unsharp masking-bilateral filter.




(a) (b)

**Fig. 20.** (a): Optimal datasets per Model and Method; (b): Distribution of optimal Re-Processing methods.

method stands out, yielding a higher ratio of 2/8 (25%) in achieving optimized accuracy. This finding suggests that certain preprocessing methods complement specific model architectures, enhancing synergistic performance. Further exploration of such synergies could unlock additional avenues for improving classification accuracy and robustness across diverse medical imaging datasets.

***Optimized Run Time***

A key aspect of our evaluation was the computational efficiency of the models. As shown in the "Average run time" chart in Fig. 15, MobileNetV2 achieves the top performance in accuracy while simultaneously exhibiting the most efficient resource usage. It boasts an average run time of just 6.97 s, whereas the second-best model, ResNet-50, has an average run time of 10.57 s. This makes MobileNetV2 the fastest among the five tested models and at least 34% faster than ResNet-50. This significant reduction in run time highlights MobileNetV2's efficiency in processing large medical imaging datasets, further solidifying its suitability for real-world applications.

***Explanation of Model Performance***

The performance of each model can be attributed to various factors, including architecture complexity, depth, and computational efficiency. Let us analyze the performance of each model in ascending order:

- *VGG16* [102]: while being a classic and widely used architecture, it suffers from its depth and parameter inefficiency. Its simplistic architecture, without Batch Normalization, comprising only 16 layers with a standard configuration of convolution and pooling, limits its ability to identify intricate patterns in the data. Moreover, the absence of skip connections or residual connections hinders gradient flow during training, leading to optimization difficulties.
- *DenseNet-169* [51]: utilizes a variety of convolutional layers, including (1,1), (3,3), and (7,7) kernels, organized into dense blocks. CNN [51] employs connected layers, where each layer is directly fed with inputs from all previous layers. This dense connectivity facilitates feature sharing and enhances gradient propagation, leading to better feature representation and model performance. Additionally, DenseNet's skip connections alleviate the vanishing gradient issue, allowing for efficient training of deeper networks.
- *ResNet-50* [103]: builds upon DenseNet's concept of skip connections but employs residual connections instead. CNN [103] uses a combination of (1,1), (3,3), and (7,7) convolutional kernels, along with skip connections. These residual connections allow the network to capture residual functions, thereby mitigating the degradation problem observed in deeper networks. By bypassing layers, ResNet-50 maintains a strong gradient flow throughout the network, facilitating easier optimization and better generalization performance.
- *MobileNetV2* [104]: introduces depthwise separable convolutions, greatly reducing the computational cost and model size, all while preserving high performance. By using inverted residual blocks, this architectural innovation enables efficient feature extraction by factorizing standard convolutions into depthwise and pointwise convolutions. As a result, MobileNetV2 achieves a balance between computational efficiency and classification performance, making it ideally suited for environments with limited resources. MobileNetV2 predominantly uses $(3 \times 3)$ convolutional layers and depthwise separable convolutions. Besides, using *ReLU6* with output limit, 6 helps reduce saturating gradients and provide a static value output range, which is useful for low-precision numerical/floating point format. Therefore, model [104] not only restricts information loss and reduces primary memory usage but also increases computation speed and keeps high accuracy.
- *EfficientNet-B4* [105]: represents cutting-edge advancements in CNN architectures, leveraging "compound scaling" to optimize model performance across multiple dimensions. By systematically scaling depth, width, and resolution, EfficientNet achieves superior performance while maintaining computational efficiency. Additionally, EfficientNet incorporates advanced architectural features such as *Swish* activations and *squeeze-and-excitation (SE)* blocks, further enhancing representational capacity and feature learning. EfficientNet-B4 utilizes a more complex architecture with a higher number of *Conv2D*, *BatchNormalization*, and *GlobalAveragePooling2D* layers, along with a mix of activation functions including *Linear*, *Sigmoid*, and *Swish*. Moreover, EfficientNet-B4 uses a larger input image size $(380 \times 380)$, allowing for better capturing of fine details in the data.

### 4.6. Comparison

Table 8 presents a comprehensive comparison of the best models and preprocessing methods applied across various datasets in medical imaging. Each dataset is analyzed alongside relevant studies, showcasing the models that achieved the highest accuracy.

## 5. Discussion

This section explores the implications of our experimental results on the four types of medical imaging datasets — radiography, Ultrasound, CT, and MRI. By evaluating the performance of different preprocessing techniques and deep learning models, we provide practical recommendations for professionals working in medical imaging and computer vision. Our analysis considers the specific characteristics of each dataset and the observed outcomes from various combinations of preprocessing methods and models.

### 5.1. X-ray datasets

Lung and knee X-ray images differ significantly in their structures and details. Lung X-rays reveal intricate features like the pulmonary network and capillaries, while knee X-rays emphasize bone structures and surrounding tissues. When diagnosing COVID-19 Pneumonia, identifying signs of infection and lung damage is crucial. In Osteoporosis, evaluating bone degradation and density changes takes precedence. Lung X-rays are generally larger and have higher resolution than knee X-rays. These variations can impact the selection and effectiveness of preprocessing methods and deep learning models.

Our experiments show a clear preference for combining Unsharp Masking and the Bilateral Filter for preprocessing. While the Median-Mean Hybrid method slightly outperforms this combination (96.47% accuracy vs. 96.32%) on Dataset 1 (Chest X-ray), the minimal difference suggests both approaches are highly effective. Unsharp Masking and the Bilateral Filter excel at highlighting important details in knee X-ray images while reducing noise. This balance of smoothing and detail preservation is crucial for the Osteoporosis Knee X-ray dataset, where accurate diagnosis depends on precisely evaluating bone loss and density changes. The Median-Mean Hybrid Filter, combining the Median and Mean filters, effectively smooths X-ray images while preserving key details. This is particularly beneficial for dataset 1 - Chest X-ray, where accurate lung health assessment requires noise reduction and retention of vital image features. Ultimately, however, the Unsharp Masking - Bilateral Filter combination stands out for its ability to enhance contrast and edge sharpness, consistently producing high-quality predictions across diverse models.

Dataset size is crucial when choosing a model. The Chest X-ray dataset's large scale enables the model to capture accurate features and reduces overfitting. Its diverse images, including variations in shape and pathologies like COVID-19, benefit from ResNet-50's deep architecture. This model excels at extracting complex information from large, varied datasets. Conversely, the Osteoporosis Knee X-ray dataset's limited size (372 images, two classes) necessitates a model capable of learning from fewer samples. DenseNet-169, with its interconnected layers, effectively learns incrementally, making it suitable for smaller, less diverse datasets.

**Table 8**
Comparison of accuracy results with related studies across datasets for different modalities.

| | Study | Best model | PreProcessing method | Best Acc. (%) |
|---|---|---|---|---|
| Dataset 1 | [106] | 12 layers CNN model | Adaptive masking, histogram equalization, and gaussian blur. | 98.31 |
| | [107] | AutoML based on two neural network architectures, ResNet and MobileNet | NA | 89.76 |
| | **This Study** | ResNet-50 | Median-Mean hybrid filter | 96.47 |
| Dataset 2 | [108] | VGG-16 Pretrain model | Grayscale conversion, data normalization, data augmentation | 86.36 |
| | [109] | DCNN - Deep Convolutional Neural Network | Mean filter, HE and Smudging Procedure | 94.70 |
| | **This Study** | DenseNet-168 | Unsharp Masking + Bilateral filter | 90.54 |
| Dataset 3 | [110] | DenseNet201 with ETC selected 1500 features | Feature scaling, label encoder | 95.00 |
| | [111] | Lung-EffNet based on EfficientNetB1 architecture | Cropping, data augmentation, label encoder | 94.00 |
| | **This Study** | EfficientNet-B4 | DWT + Threshold | 87.62 |
| Dataset 4 | [112] | BiLSTM model | Resize | 96.50 |
| | [113] | Xception with Linear Discriminant classifiers | Data augmentation | 98.60 |
| | **This Study** | MobileNetV2 | Median-Mean hybrid filter | 96.20 |
| Dataset 5 | [114] | NeuroNet19 based on VGG19 architecture | Grayscale conversion, gaussian blur, otsu threshold, cropping. | 99.30 |
| | [115] | MobileNetv3 | Data augmentation | 99.75 |
| | **This Study** | EfficientNet-B4 | Median-Mean hybrid filter | 98.55 |
| Dataset 6 | [117] | ResNet-50 | Resize, data augmentation | 92.01 |
| | [116] | AlexNet | Resize | 98.10 |
| | **This Study** | MobileNetV2 | Unsharp Masking + Bilateral filter | 83.75 |
| Dataset 7 | [118] | VGG19 | Data augmentation, sharpening | 98.44 |
| | [119] | MobileNetV2 Plus (Wavelet Convolutional Neural Network) | Normalization, sharpening, data augmentation | 92.78 |
| | **This Study** | EfficientNet-B4 | Unsharp Masking + Bilateral filter | 98.78 |
| Dataset 8 | [22] | DeepLabV3+-ResNet50 | Fuzzy Intensification Operator (FIO) | 96.15 |
| | [120] | ResNet50-Adagrad-based MSTL | Data augmentation, normalization | 98.70 |
| | **This Study** | EfficientNet-B4 | Unsharp Masking + Bilateral filter | 96.25 |

### 5.2. CT datasets

For CT datasets, our research underscores the effectiveness of two preprocessing methods: DWT thresholding and the Median-Mean Hybrid Filter, each suited to specific dataset complexities. Chest and brain CT images pose distinct challenges due to their unique structures and diagnostic requirements.

Chest CT scans featuring intricate details such as blood vessels, lungs, and ribs demand preprocessing methods that enhance fine details while minimizing noise. DWT + Thresholding excels in this regard, achieving an accuracy of 87.62% by accentuating internal features and reducing noise effectively. The discrete wavelet transform (DWT) plays a pivotal role here, adeptly filtering noise while preserving critical image details. However, the choice of wavelet basis and thresholding approach is critical to strike an optimal compromise between minimizing noise and preserving details.

Conversely, brain stroke CT images prioritize the detection of stroke indicators, such as bright areas and density changes. The Median-Mean Hybrid Filter, which combines smoothing with the preservation of crucial image features, proves highly effective in this context, achieving an impressive 96.20% accuracy. This method smoothes images while retaining essential details vital for accurate stroke detection, underscoring its significance in identifying and diagnosing these critical conditions.

Given the complex structures in chest CT images and the dataset's four classes, complexity in classification surpasses that of brain stroke CT images, which have only two classes. EfficientNet-B4, with its advanced network architecture featuring MBConv1 and MBConv6 blocks, excels at learning complex features necessary for distinguishing multiple classes. Its increased parameter count and architectural complexity enable EfficientNet-B4 to generalize effectively and prevent overfitting or underfitting, making it adept at handling such intricacies in classification tasks. Conversely, brain CT images, focusing on detecting stroke manifestations, pose simpler challenges in classification. MobileNetV2, with its lightweight and efficient design optimized for mobile devices, proves proficient at handling less complex tasks effectively.

### 5.3. MRI datasets

In the evaluated MRI datasets, the Unsharp Masking + Bilateral Filter method consistently demonstrates superior effectiveness.

Dataset 5 — Brain MRI, containing 7023 images and a complex 4-class classification task, presents a challenging scenario. The intricate details within these images, coupled with the need to define tumor boundaries for accurate classification clearly, necessitate robust image processing. The Median-Mean Hybrid Filter, achieving a top accuracy of 98.55%, excels by effectively combining median and mean filtering, reducing noise while preserving edges. Brain tumor MRI images often exhibit high noise levels and subtle intensity variations, which are crucial for precise classification. This hybrid approach ensures that important features, like tumor boundaries, are enhanced while noise is minimized, resulting in high accuracy. However, Unsharp Masking + Bilateral Filter also achieved very impressive results, reaching an accuracy of 98.47%. Unsharp masking enhances edges by emphasizing high-frequency components, while the bilateral filter reduces noise while preserving edges. This combination effectively highlights important structural details in the MRI images, such as tumor edges and

texture, which are crucial for distinguishing between different classes. The slight drop in accuracy compared to the Median-Mean Hybrid Filter may be attributed to the over-enhancement of certain features or residual noise. Given the complex data characteristics and multi-class classification task, the EfficientNet-B4 model again achieved the highest classification efficiency on this dataset. EfficientNet-B4 is known for its sophisticated architecture that balances depth, width, and resolution in a highly efficient manner. The network's use of compound scaling and MBConv blocks enables it to discern intricate patterns and features within the MRI images effectively. The higher number of parameters and layers enables EfficientNet-B4 to excel in learning from the more complex, multi-class data present in the Brain Tumor MRI dataset, leading to superior performance.

Dataset 6 — Breast MRI consists of 1480 images with two classes. Due to its lower complexity and binary classification task, specific noise reduction and edge enhancement methods can yield good performance. An approach combining Unsharp Masking and Bilateral Filtering demonstrated remarkable outcomes with this dataset. On the other hand, given the characteristics of this dataset, MobileNetV2 emerged as the most suitable model due to its lightweight design. MobileNetV2, equipped with depthwise separable convolutions and bottleneck layers, proved to be effective in the binary classification task, becoming a fitting selection for dataset 6.

### 5.4. Ultrasound datasets

We conducted comprehensive experiments on two ultrasound imaging datasets related to breast cancer, each consisting of two classification classes. The combination of EfficientNet-B4 and the Unsharp Masking + Bilateral Filter pre-processing method achieved outstanding performance on both ultrasound datasets. The model's powerful deep learning capabilities, paired with the enhanced image quality provided by the pre-processing method, led to more accurate and reliable classification results. This combination demonstrated a marked enhancement in identifying and classifying breast cancer in ultrasound scans, overcoming the typical challenges of noise and low contrast.

Ultrasound images are inherently challenging due to significant noise and low contrast. These images often suffer from artifacts caused by the equipment and varying light conditions during the imaging process. The noise and low contrast obscure important features, making it difficult for models to accurately identify and classify abnormalities.

The Unsharp Masking + Bilateral Filter pre-processing method improves image quality by enhancing important features and reducing noise. Unsharp Masking increases contrast around edges to highlight details, while the Bilateral Filter reduces noise while preserving essential boundaries and structures within the images. This is particularly useful in ultrasound imaging, where light conditions and equipment noise can affect image quality. Unsharp Masking enhances the images by subtracting a smoothed version from the original, which sharpens the image and makes the edges more prominent. This enhancement is crucial for highlighting important features that are otherwise difficult to discern in noisy ultrasound images. Unlike traditional smoothing filters that might blur edges, the Bilateral Filter smooths the image while maintaining the sharpness of edges, making it particularly suitable for medical imaging where preserving structural integrity is vital. This filter is especially effective in environments with inconsistent light conditions and equipment noise typical of ultrasound imaging.

EfficientNet-B4 excels in the classification of ultrasound data due to its powerful DL capabilities. When paired with carefully preprocessed input images, EfficientNet-B4 effectively learns crucial features, resulting in highly accurate classification outcomes. Its well-balanced architecture optimizes parameters, depth, and width, which proves instrumental in achieving robust generalization across ultrasound image datasets. This exceptional generalization capability is a critical factor in EfficientNet-B4's standout performance in our experiments. The enhanced clarity of images from the Unsharp Masking + Bilateral Filter preprocessing method equips the model with distinctly clearer features. This clarity facilitates improved learning, thereby enhancing the model's accuracy in classification tasks.

### 5.5. Recommendations for practitioners

Based on our extensive experimental results, we provide the following detailed recommendations for professionals in the fields of medical imaging and computer vision. These insights are intended to help practitioners optimize their workflows, enhance diagnostic accuracy, and improve patient outcomes by selecting the most effective preprocessing methods and deep learning models tailored to specific imaging modalities.

(1) **X-ray Datasets:**

- **Preprocessing:** Utilize Unsharp Masking to enhance image contrast and edge definition and Bilateral Filter to reduce noise while preserving important structural details.
- **Model Selection:** For large datasets, opt for ResNet50 due to its deep architecture and extensive feature extraction capabilities. For smaller datasets, select DenseNet-169 for its efficient learning from limited data.

(2) **CT Datasets:**

- **Preprocessing:** Employ DWT thresholding for moderate complexity datasets to balance noise reduction with the preservation of critical internal features. Use the Median-Mean Hybrid Filter for highly complex datasets to smooth while retaining key diagnostic features.
- **Model Selection:** Choose MobileNetV2 for fewer class datasets due to its lightweight design and efficiency. For multi-class datasets, select EfficientNet-B4 for its balanced architecture that handles complexity well.

(3) **MRI Datasets:**

- **Preprocessing:** Apply Unsharp Masking to enhance subtle details and improve the visibility of fine structures.
- **Model Selection:** For fewer class datasets, MobileNetV2 is preferred due to its efficiency. For multi-class datasets, EfficientNet-B4 is optimal for managing and learning from complex data.

(4) **Ultrasound Datasets:**

- **Preprocessing:** Use Unsharp Masking and Bilateral Filtering to enhance edges and reduce noise, improving the quality of ultrasound images.
- **Model Selection:** EfficientNet-B4 is recommended for its powerful deep learning capabilities and balanced architecture, suitable for the complexities of ultrasound imaging.

These recommendations are designed to guide practitioners in making informed decisions regarding the preprocessing and modeling techniques most appropriate for their specific medical imaging tasks. By adopting these best practices, professionals can enhance the diagnostic accuracy and efficiency of their image analysis processes, ultimately leading to enhanced patient care and better health outcomes.

### 5.6. Managerial implications

The findings from this study offer several important managerial implications for healthcare institutions, research centers, and medical imaging practitioners, particularly in the field of diagnostic image analysis. The integration of advanced deep learning models with effective preprocessing techniques can significantly enhance diagnostic accuracy, which is critical for improving patient outcomes and optimizing healthcare workflows.

Healthcare institutions aiming to implement AI-based diagnostic tools should focus on tailoring their preprocessing pipelines to the

specific imaging modality and the diagnostic task at hand. Our results demonstrate that different combinations of image enhancement techniques (e.g., Unsharp Masking, Bilateral Filtering) significantly influence model performance across datasets, thus emphasizing the importance of aligning preprocessing methods with the unique characteristics of medical imaging data, such as X-ray or MRI scans. Efficient preprocessing can reduce the need for clinicians to perform large-scale manual interventions, facilitating faster, more reliable diagnostics.

Additionally, the choice of the DL model is pivotal in resource management. For example, models like EfficientNet-B4 offer superior performance across multiple imaging modalities but come with higher computational costs. In contrast, lightweight models like MobileNetV2 can be deployed in resource-constrained environments without a significant loss in accuracy. Managers overseeing budget allocation for AI infrastructure must consider both the computational resources and the specific diagnostic needs when selecting models for implementation.

By combining preprocessing techniques with appropriate model selection, healthcare administrators can further improve the efficiency of diagnostic systems. This approach can help hospitals and research centers maximize their computational resources, reduce system downtime, and increase diagnostic throughput. For medical staff, having precise and efficient diagnostic tools enhances their ability to make timely decisions, ultimately improving patient care and healthcare management. The broader implications for the healthcare industry include the potential to integrate AI-enhanced diagnostic systems into large-scale IoT networks, where real-time data analysis could be used to monitor patient conditions remotely. Efficient AI-based systems can significantly reduce the strain on healthcare professionals, streamline diagnostic procedures, and contribute to the broader digital transformation effort in healthcare.

## 6. Conclusion

***Findings:***

This study systematically evaluated the effectiveness of five prevalent image preprocessing techniques combined with five widely recognized DL models for diagnostic purposes across various medical image datasets. By exploring combinations of CLAHE + Butterworth, DWT + Threshold, CLAHE + Median Filter, Median-Mean Hybrid Filter, and Unsharp Masking + Bilateral Filter with EfficientNet-B4, ResNet-50, DenseNet-169, VGG16, and MobileNetV2, we aimed to determine optimal configurations for different imaging modalities, including X-ray, MRI, CT-scan, and ultrasound images.

Preprocessing plays a crucial role in model performance, with combinations like unsharp masking and bilateral filtering proving highly effective. These techniques enhance image sharpness while preserving essential features, improving classification accuracy. The median-mean hybrid filter also demonstrates significant enhancements, emphasizing the importance of balancing noise reduction and feature preservation through hybrid approaches.

EfficientNet-B4 and MobileNetV2 stand out as top performers across various datasets. EfficientNet-B4, leveraging compound scaling and advanced architectural features, consistently achieves high accuracy, particularly excelling in ultrasound image classification. Meanwhile, MobileNetV2 showcases versatility and efficiency, delivering competitive performance with minimal computational resources, which is ideal for resource-constrained environments. The synergy observed between EfficientNet-B4 and preprocessing techniques like unsharp masking, and bilateral filtering suggests significant potential for further performance improvements, highlighting the importance of selecting tailored preprocessing techniques and model architectures for medical imaging tasks.

***Research Limitations:***

Despite the comprehensive evaluation, this study has certain limitations. Firstly, the scope was confined to a set of five preprocessing techniques and five DL models, which, while significant, do not encompass the full spectrum of available methods and architectures. Additionally, the study was conducted on a limited number of datasets representing four imaging modalities; thus, the findings may not generalize across all types of medical images or newer, more complex datasets.

Moreover, the evaluation of preprocessing techniques was performed uniformly across datasets without adapting techniques to the specific characteristics of each dataset. This could result in suboptimal preprocessing for certain datasets, and a more dynamic, dataset-specific approach could yield different results. The resource efficiency assessment was based on standard configurations without extensive optimization, which may affect the perceived efficiency of some models.

***Future Work:***

Future research should explore a broader range of preprocessing techniques and DL models, including emerging methods and architectures that are continually evolving in the field of medical image analysis. Investigating adaptive preprocessing strategies that dynamically adjust to the unique characteristics of each dataset could provide more nuanced insights and further enhance model performance. Additionally, expanding the evaluation to include a more comprehensive array of imaging modalities, such as PET scans or hybrid imaging techniques, would strengthen the generalizability of the findings. It is also essential to consider the integration of domain-specific knowledge into preprocessing and model selection, tailoring these processes to the clinical context.

Future studies should place greater emphasis on evaluating the resource efficiency of DL models and their performance, particularly in real-world clinical settings where computational resources are often limited. This approach would support the practical deployment of DL models in healthcare, facilitating the adoption of advanced imaging analysis techniques in expanding IoT networks and enhancing overall diagnostic capabilities.

## CRediT authorship contribution statement

**Thien B. Nguyen-Tat:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Tran Quang Hung:** Software, Writing – original draft, Visualization. **Pham Tien Nam:** Software, Writing – original draft, Visualization. **Vuong M. Ngo:** Conceptualization, Writing – review & editing.

## Data and code availability

The datasets and codes utilized and developed during this study are available from the corresponding author upon reasonable request.

## Ethics approval

Not applicable. This study did not involve any medical tests, treatments, or interventions.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The datasets and codes utilized and developed during this study are available from the corresponding author upon reasonable request.

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